

An architectural rendering of a modern building project. On the left is a traditional building with a large, ornate, vaulted facade and arched windows. In the center is a new, multi-story glass structure with a grid-like frame, revealing interior spaces with plants and people. To the right is a modern building with a curved facade and horizontal bands. The scene is set in a courtyard with people walking and sitting under umbrellas, suggesting a rainy day.

Vault House

Yiran Li

Erin Wai In Wong

Sebastián Canovas Quintana

Vaults in Heidelberg

Altes Hallenbad and Heiliggeistkirche



Structurally Optimized 3D Printed Vaulted Slabs

Philipp Block Research Group: Rippmann Slabs



Objectives of the Slabs

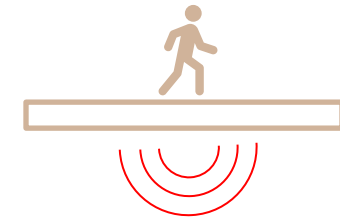
Structure



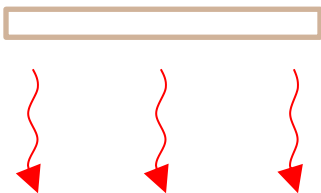
Airborne Noise



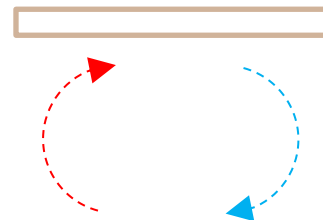
Structure-Borne Noise



Thermal Radiation



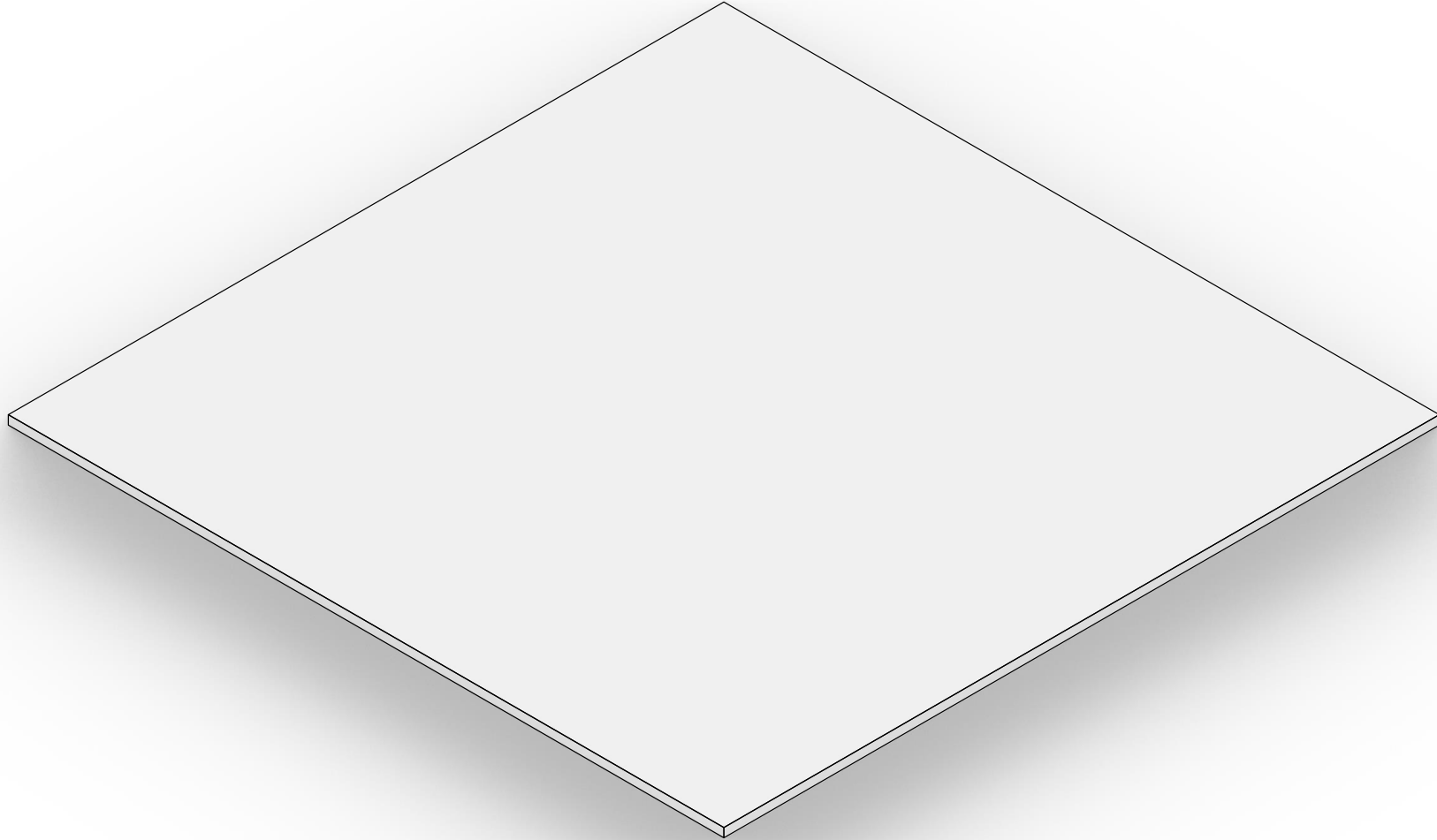
Convection



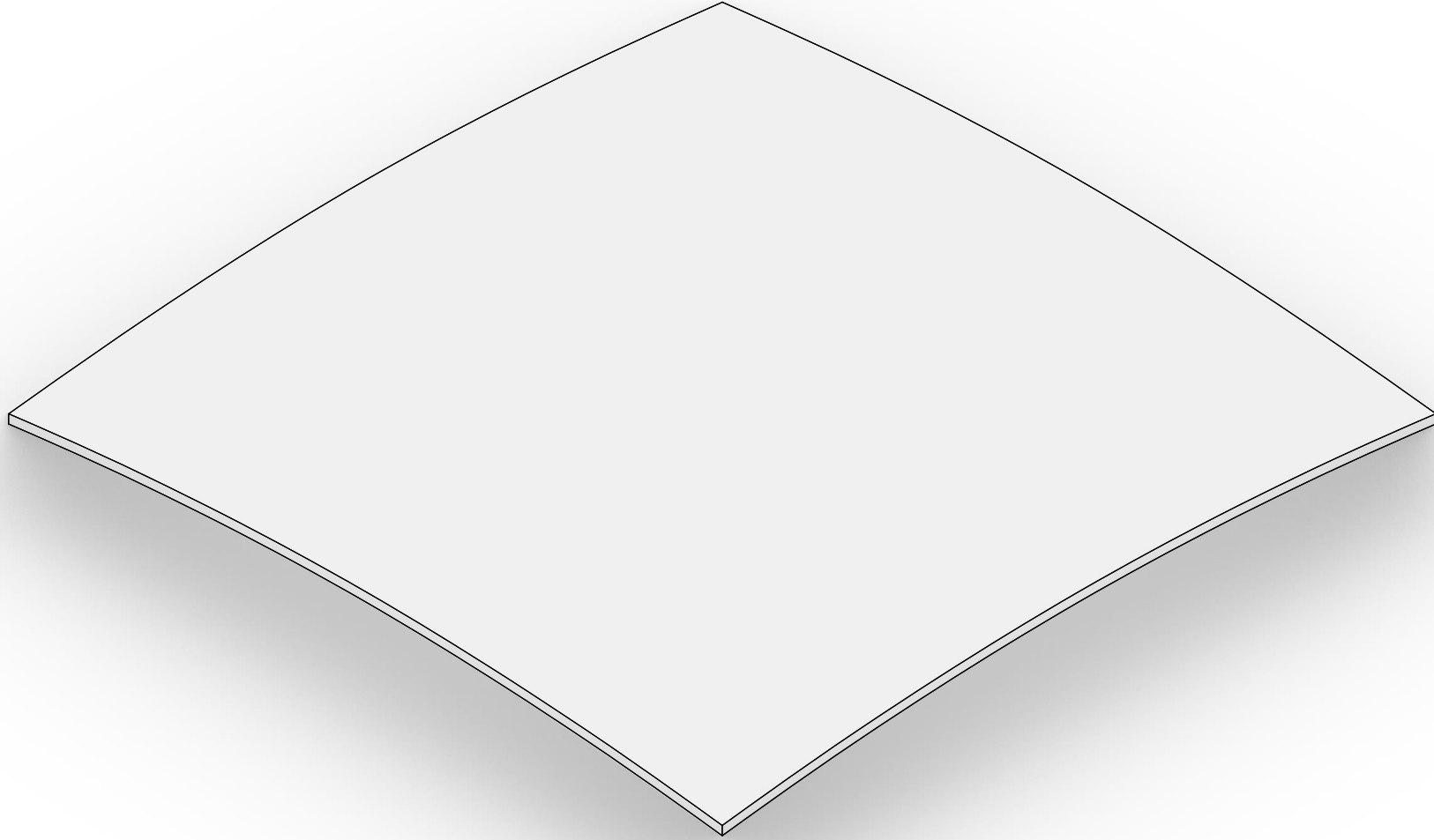
Aesthetics



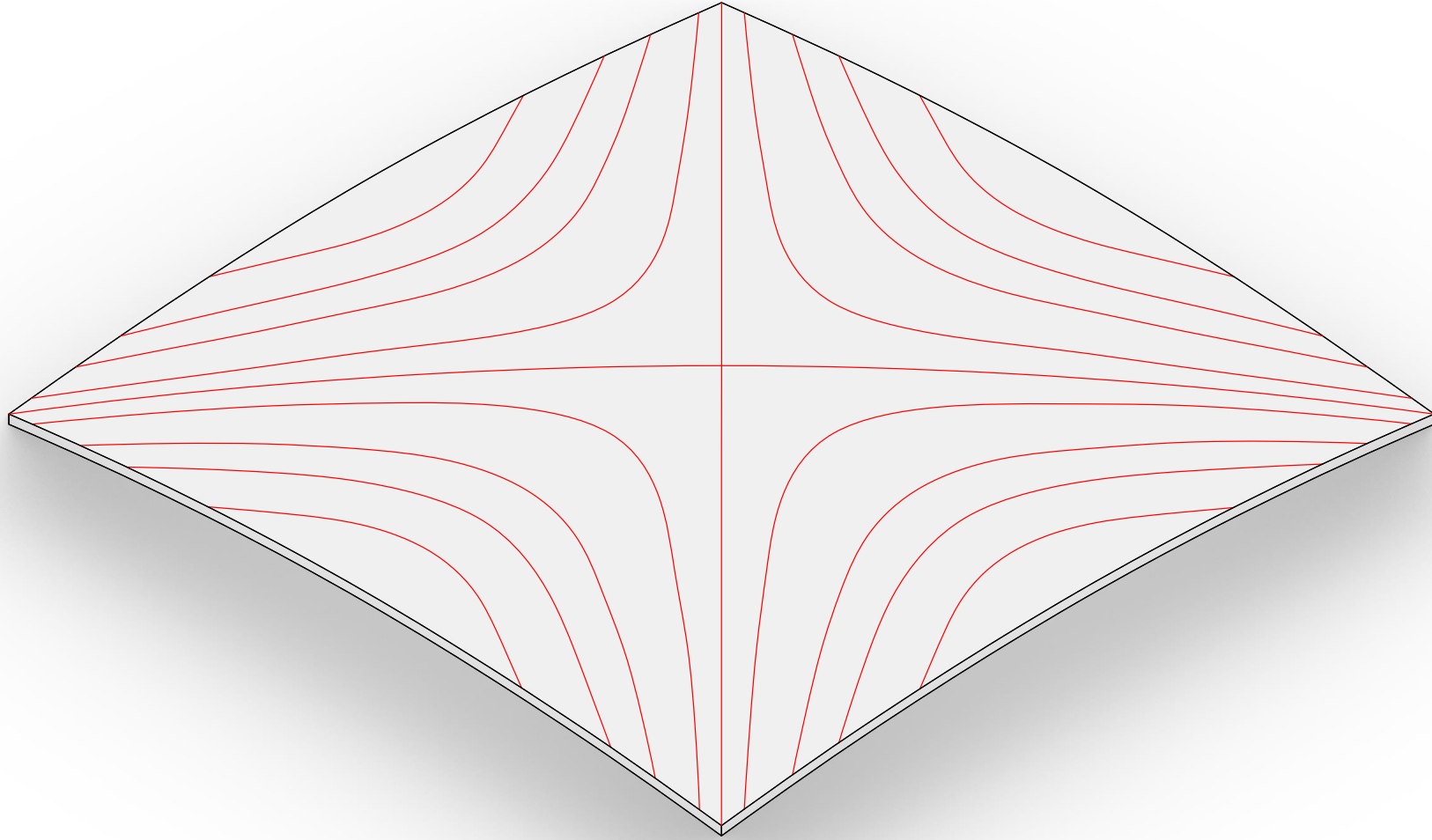
Development of the Slabs



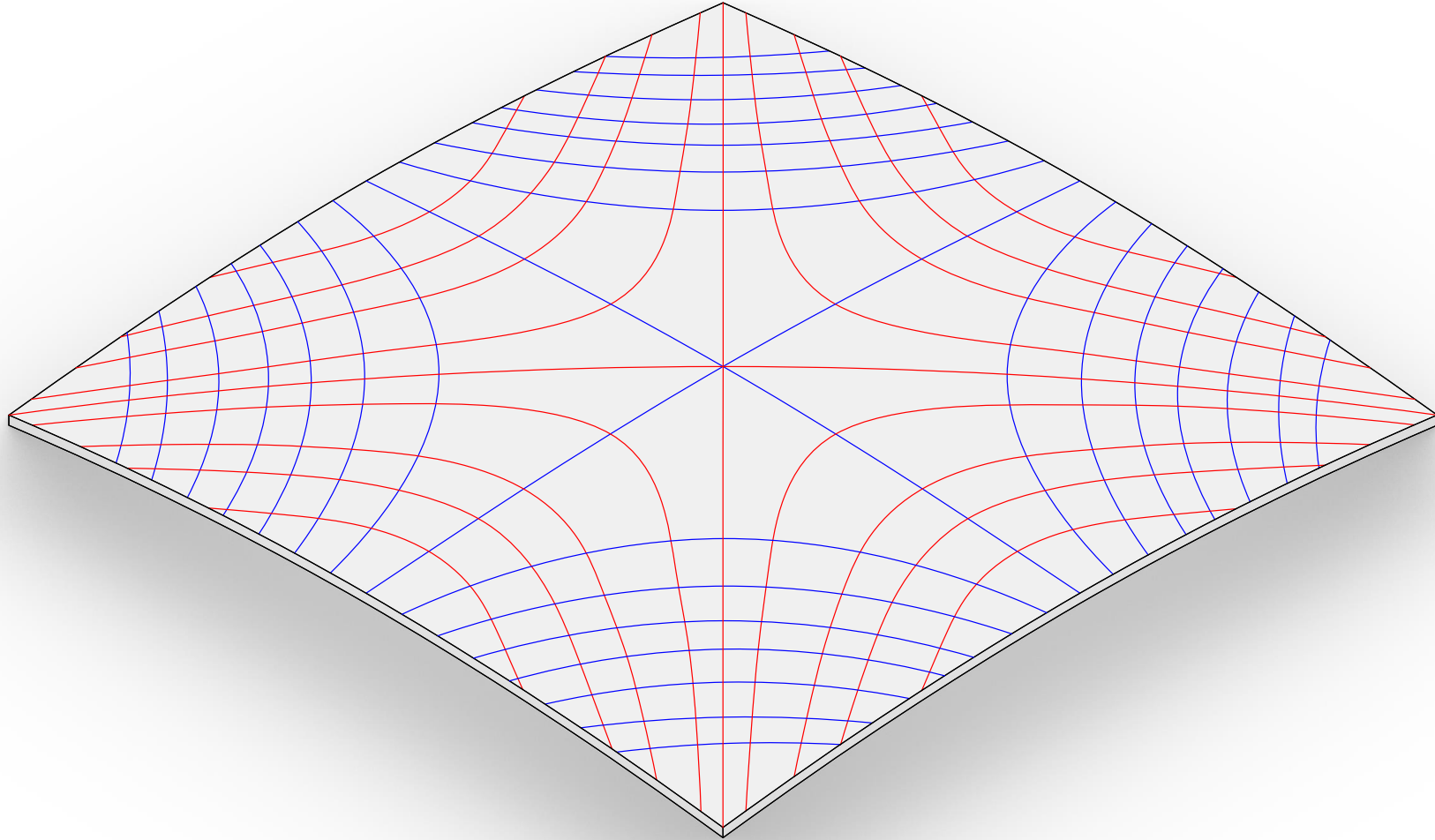
Development of the Slabs



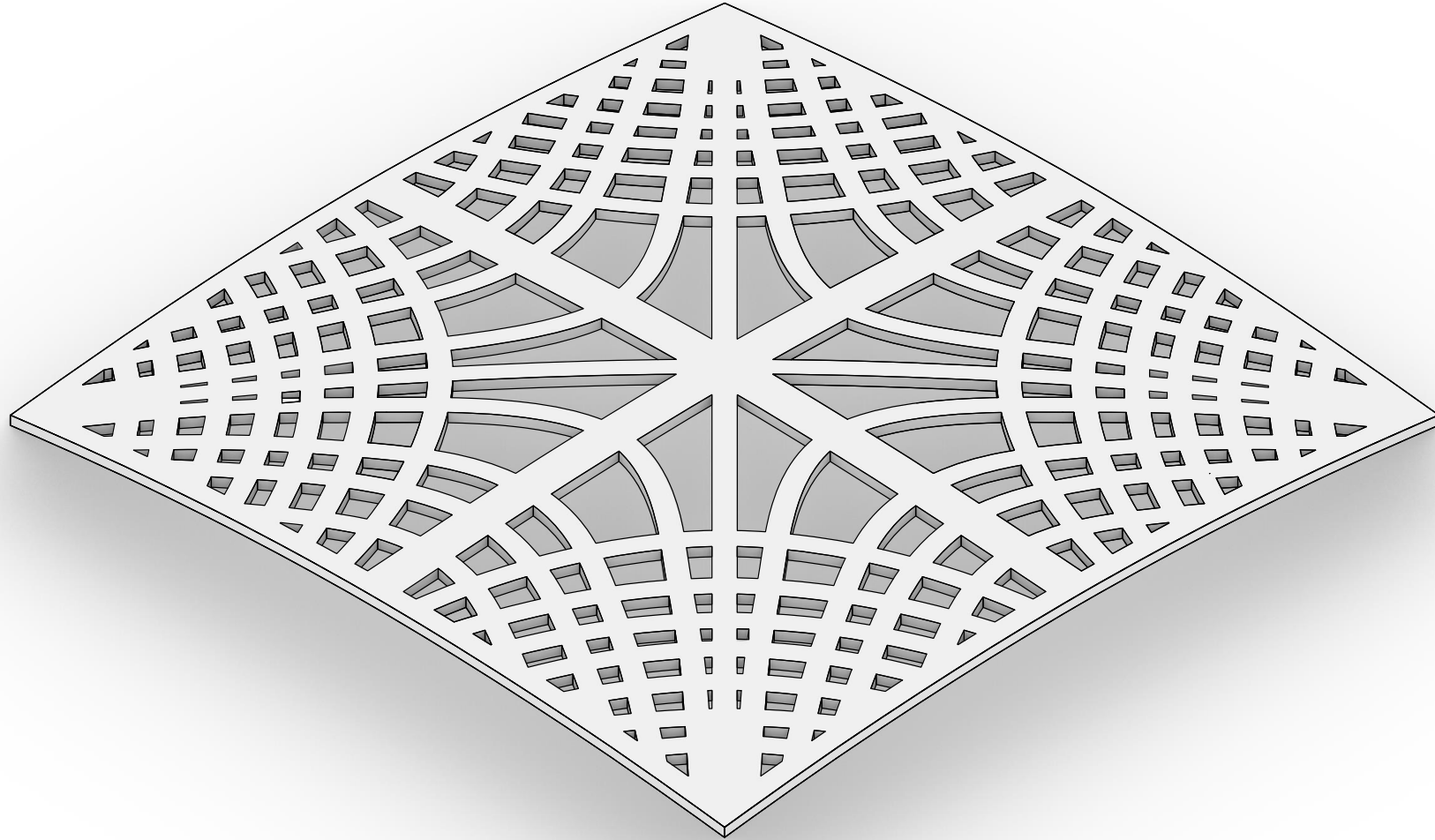
Development of the Slabs



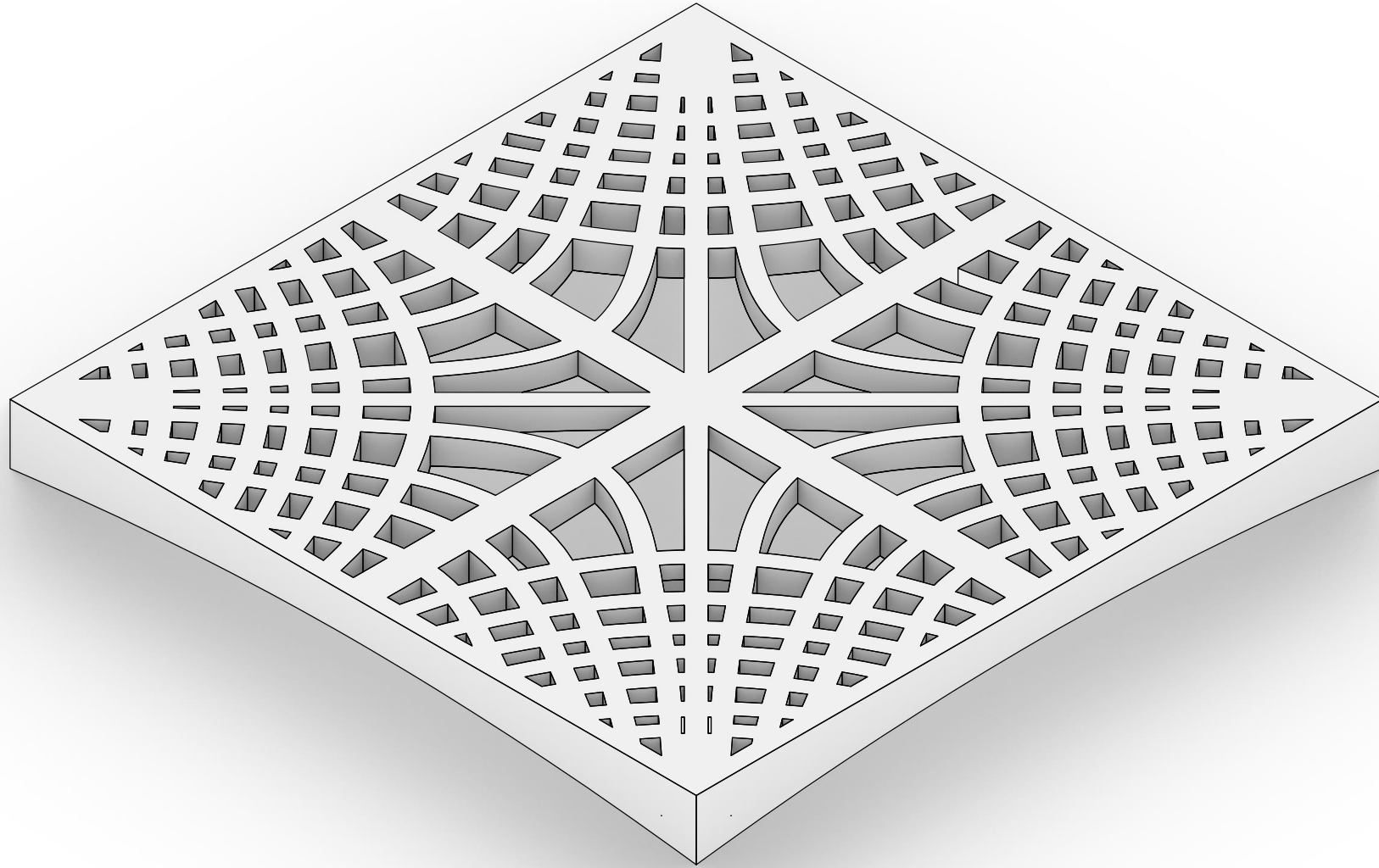
Development of the Slabs



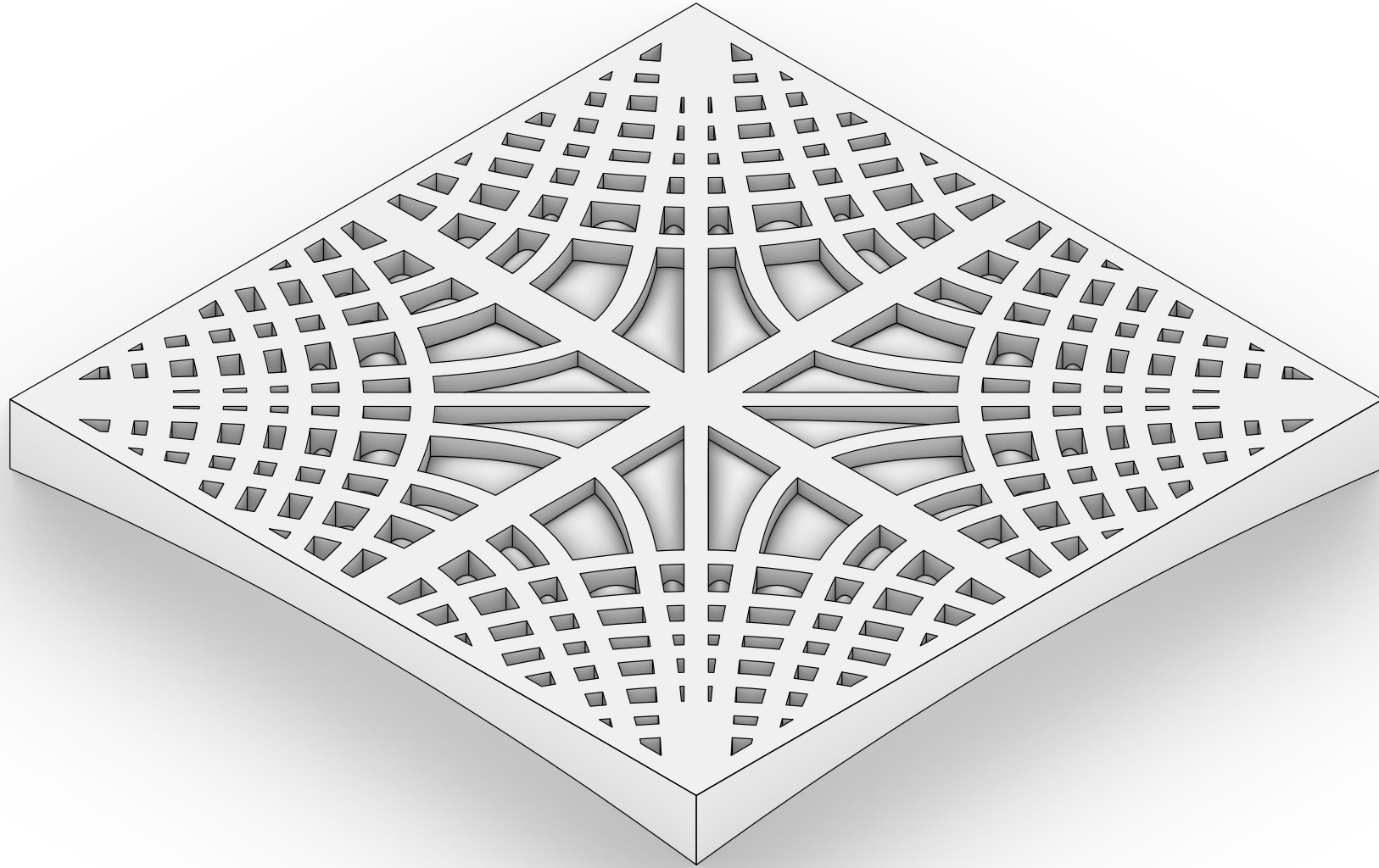
Development of the Slabs



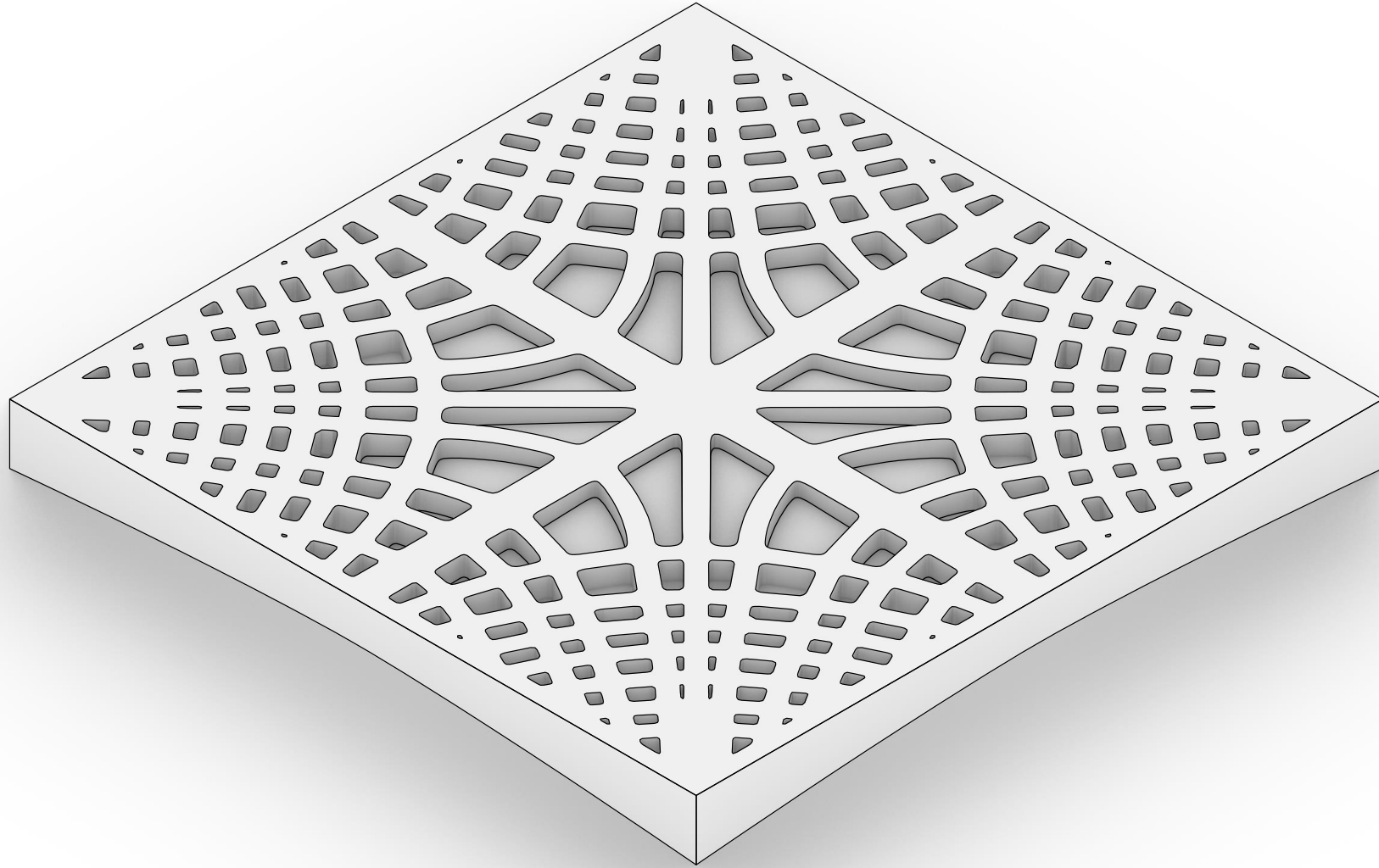
Development of the Slabs



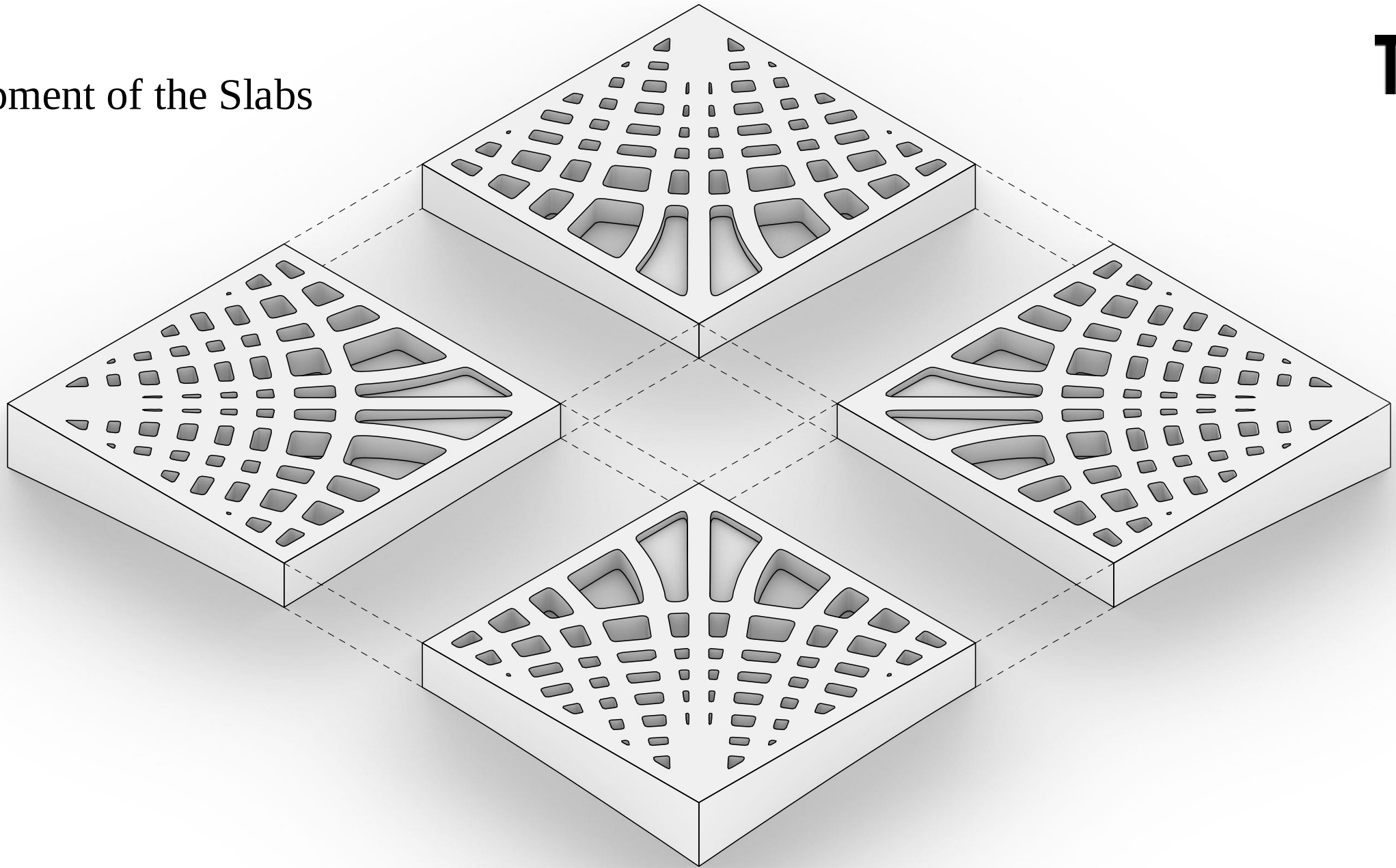
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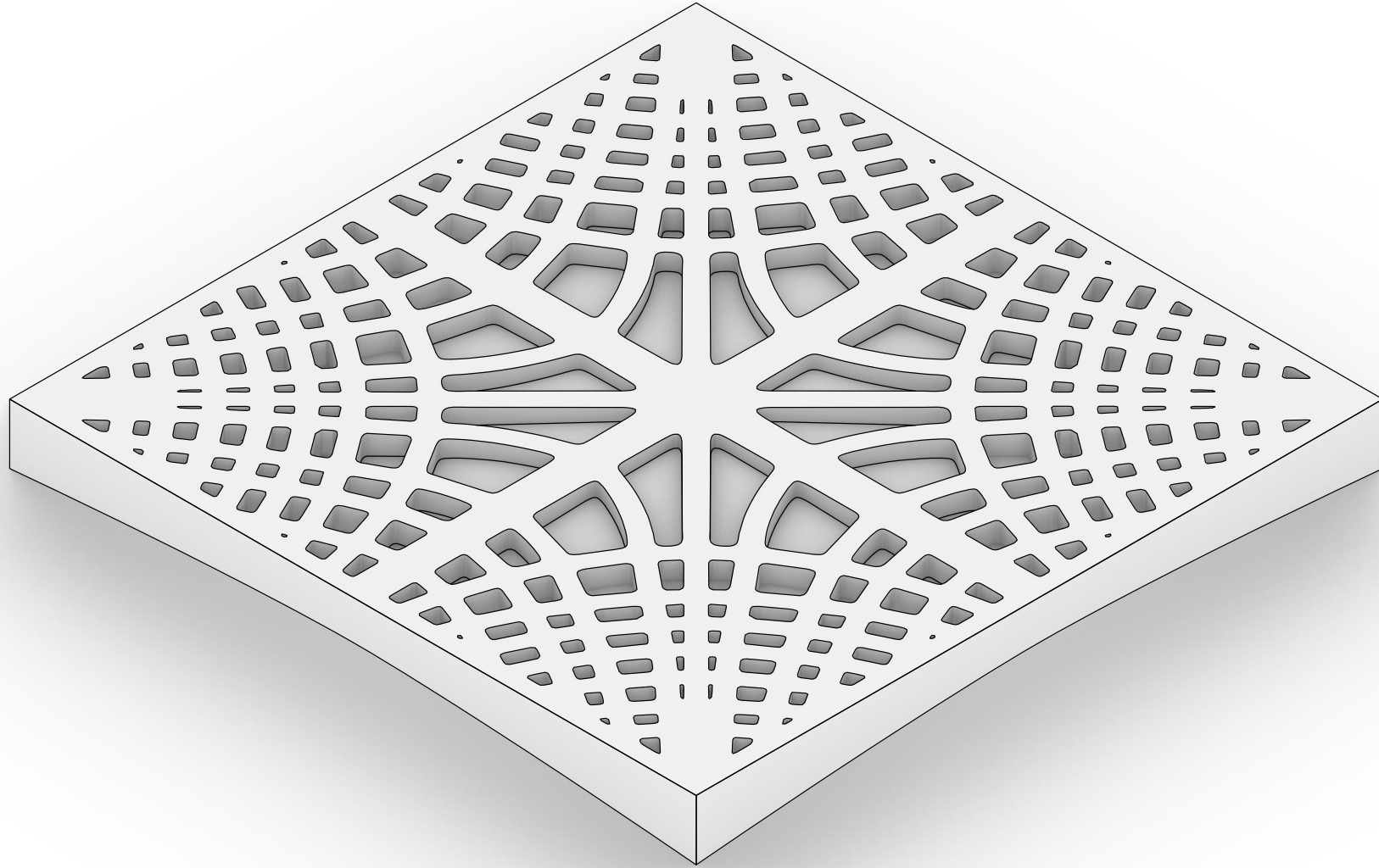
Development of the Slabs



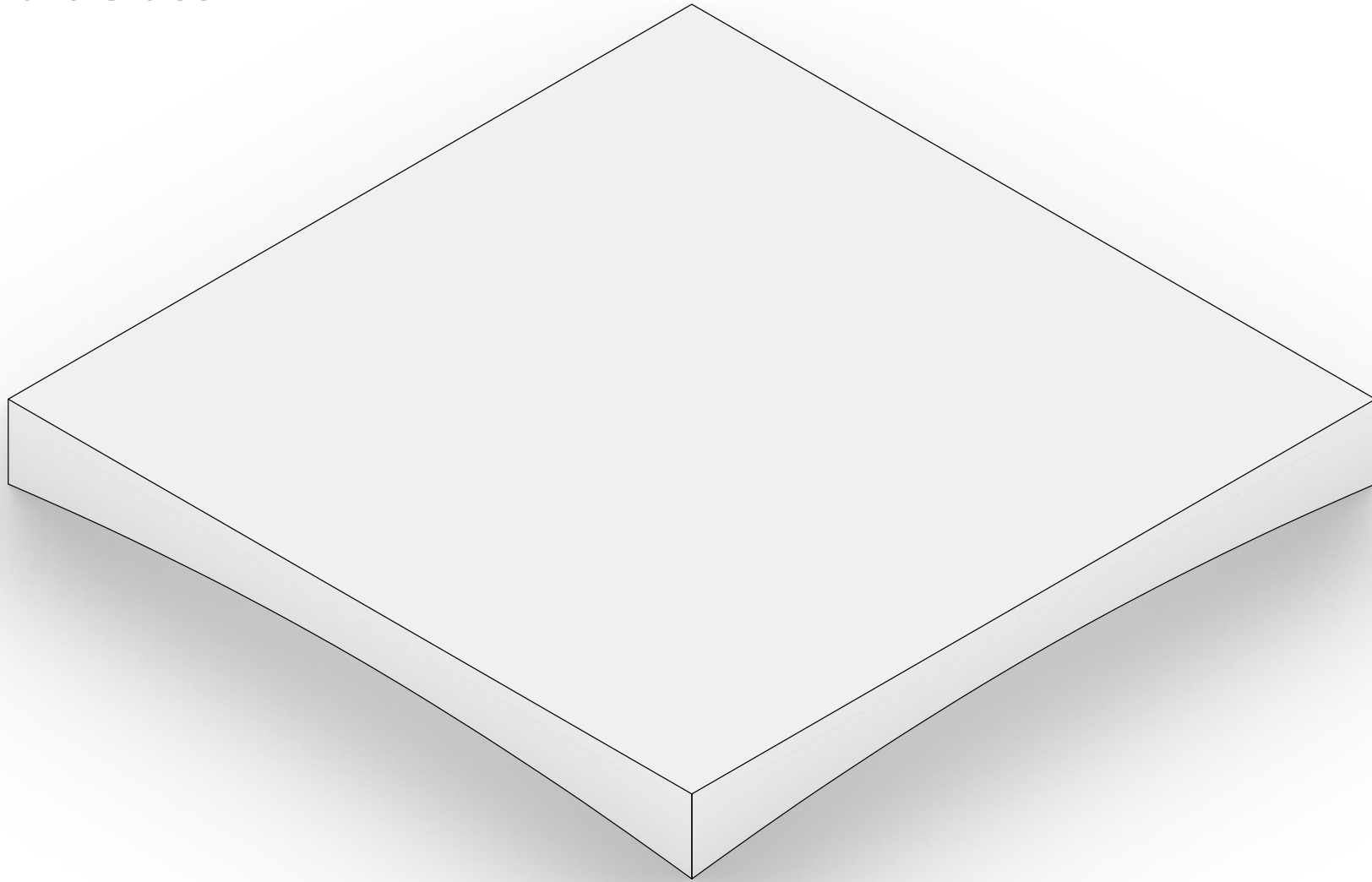
Development of the Slabs



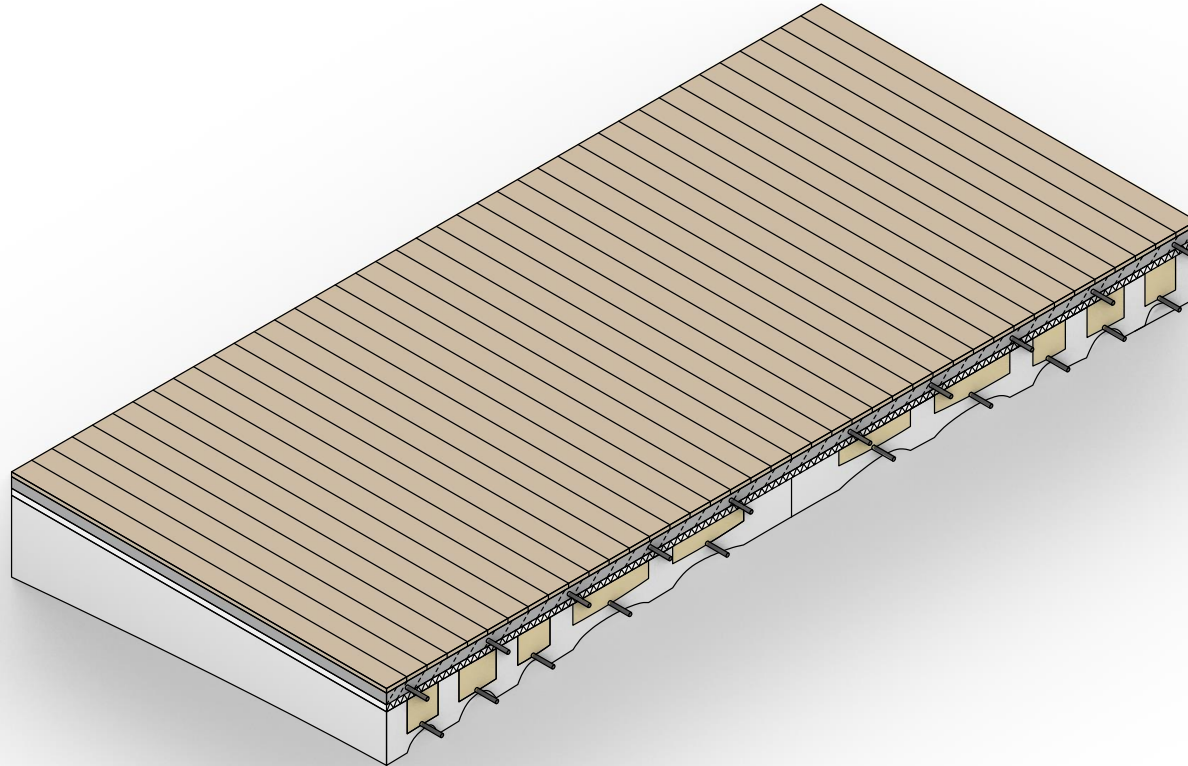
Development of the Slabs



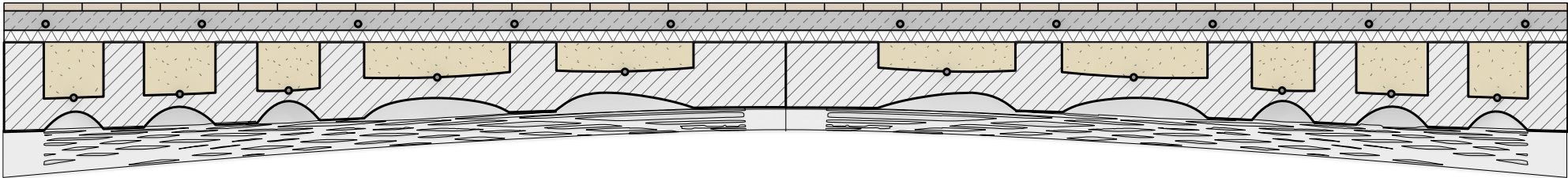
Development of the Slabs

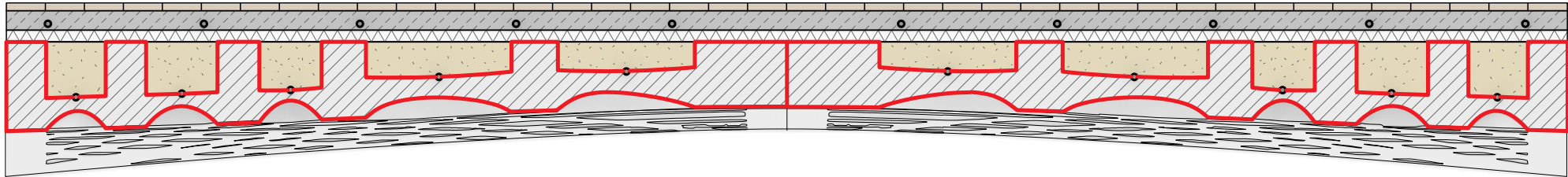


Section

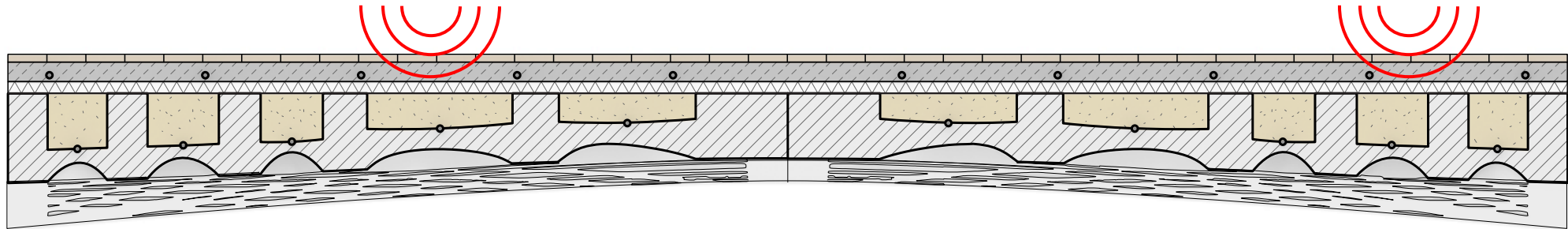


Section

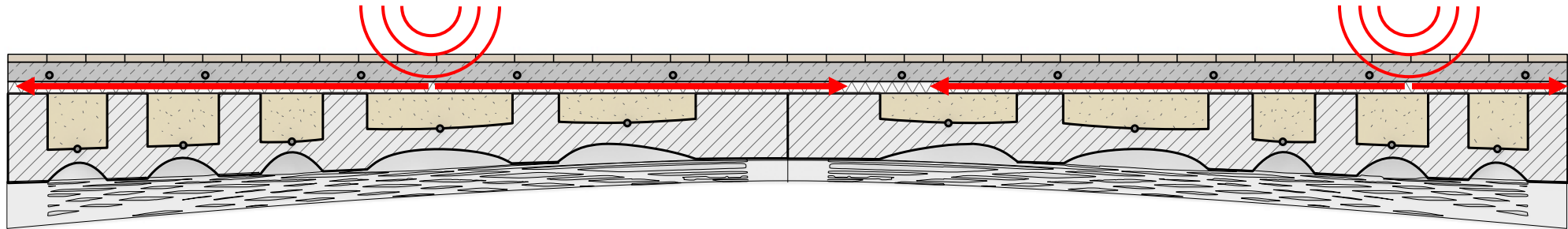




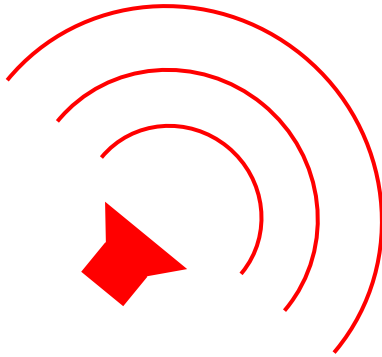
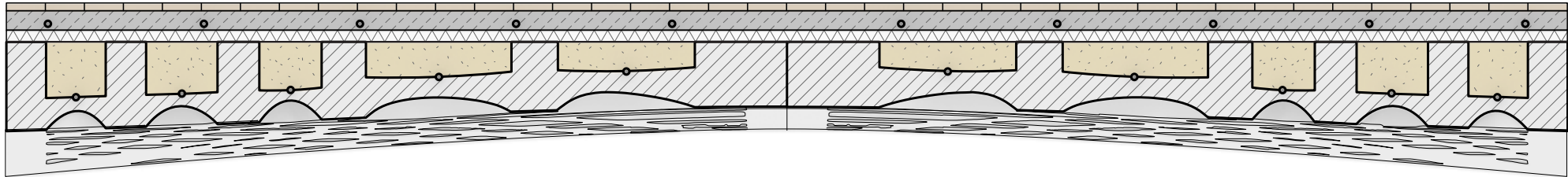
Structure-Borne Noise



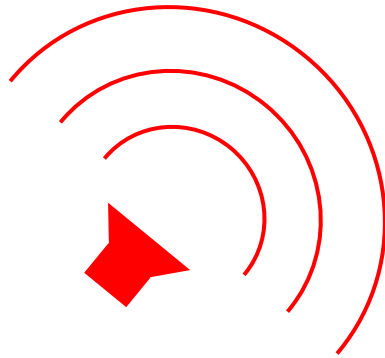
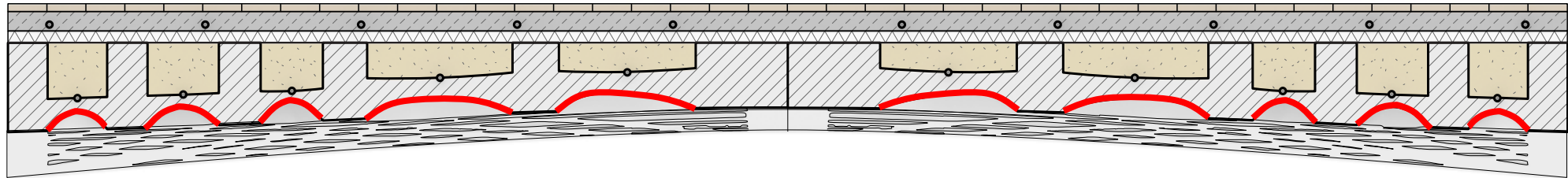
Structure-Borne Noise: Impact Sound Insulation



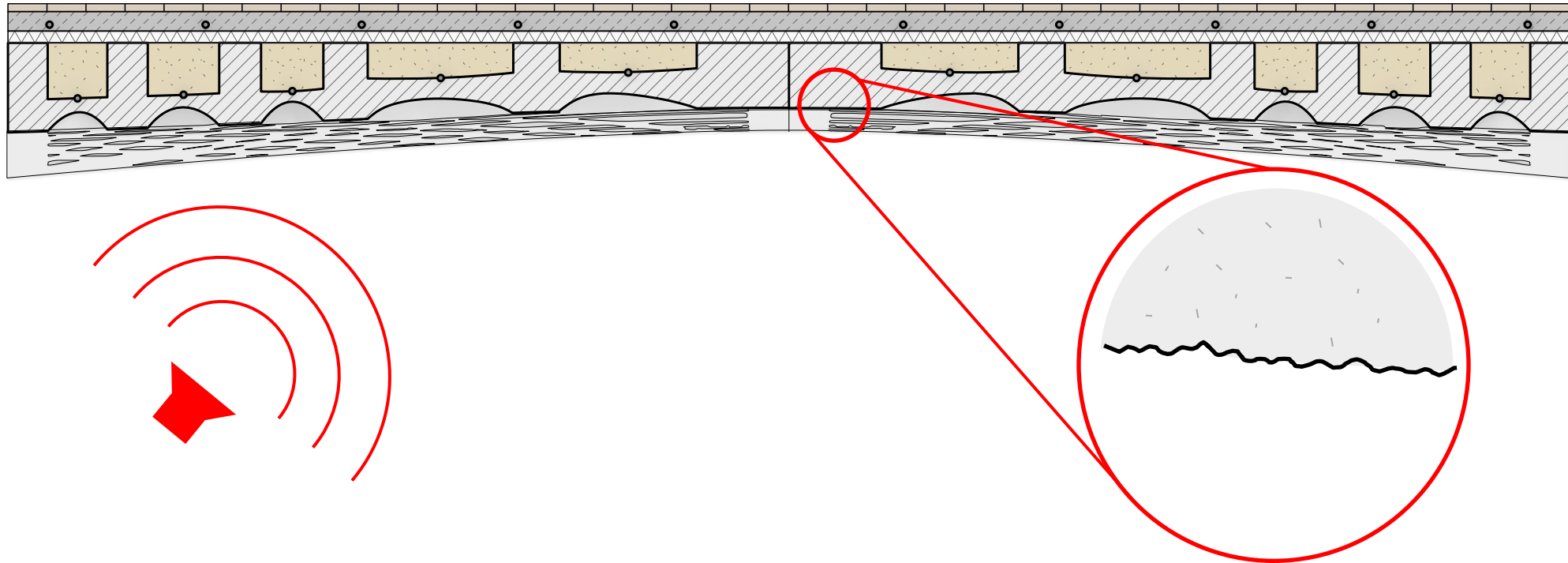
Air-Borne Noise



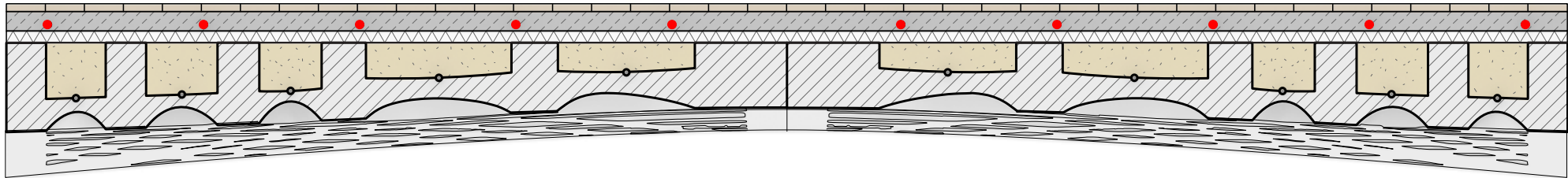
Air-Borne Noise: Surface Meso-Structure



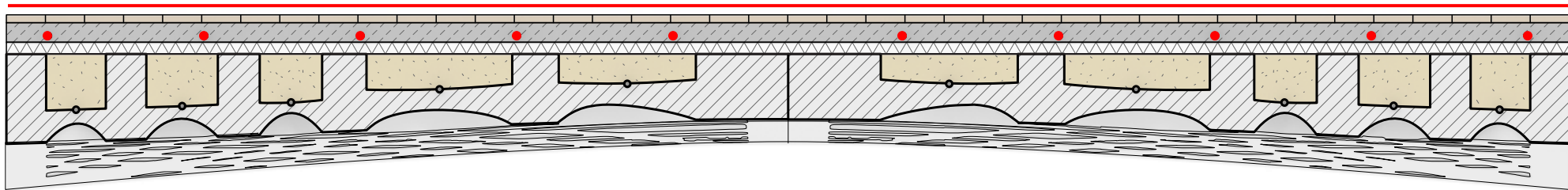
Air-Borne Noise: Surface Micro-Structure



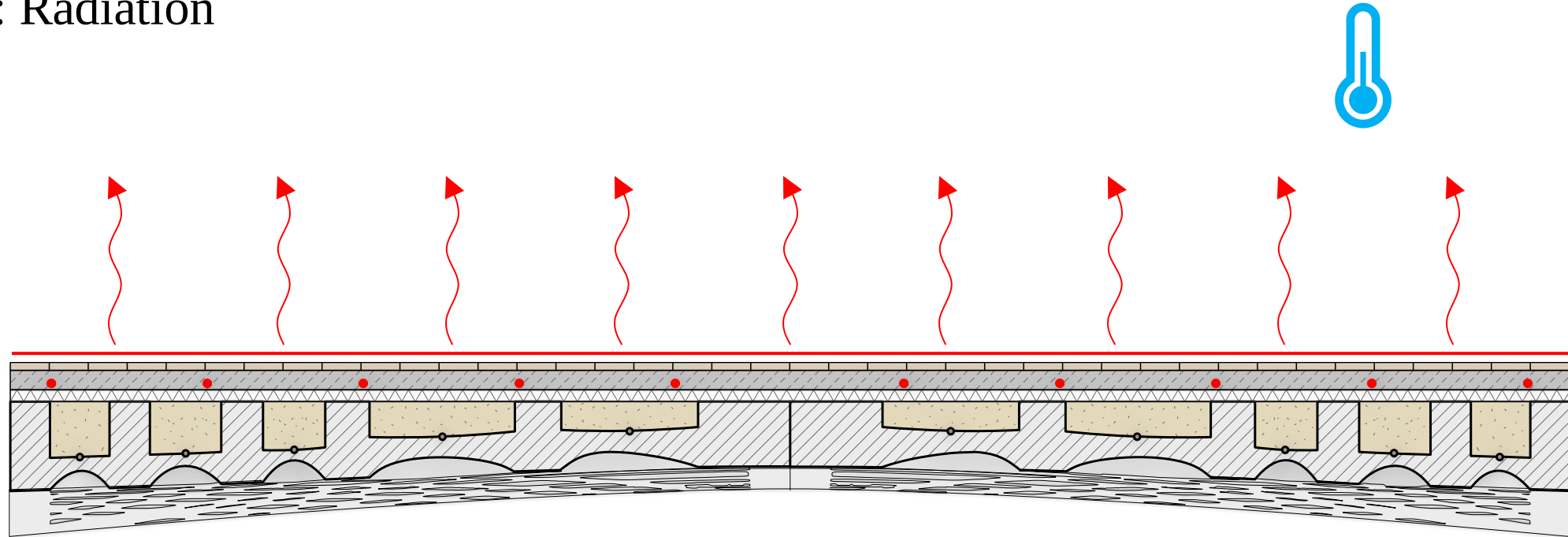
Heating: Floor Heating



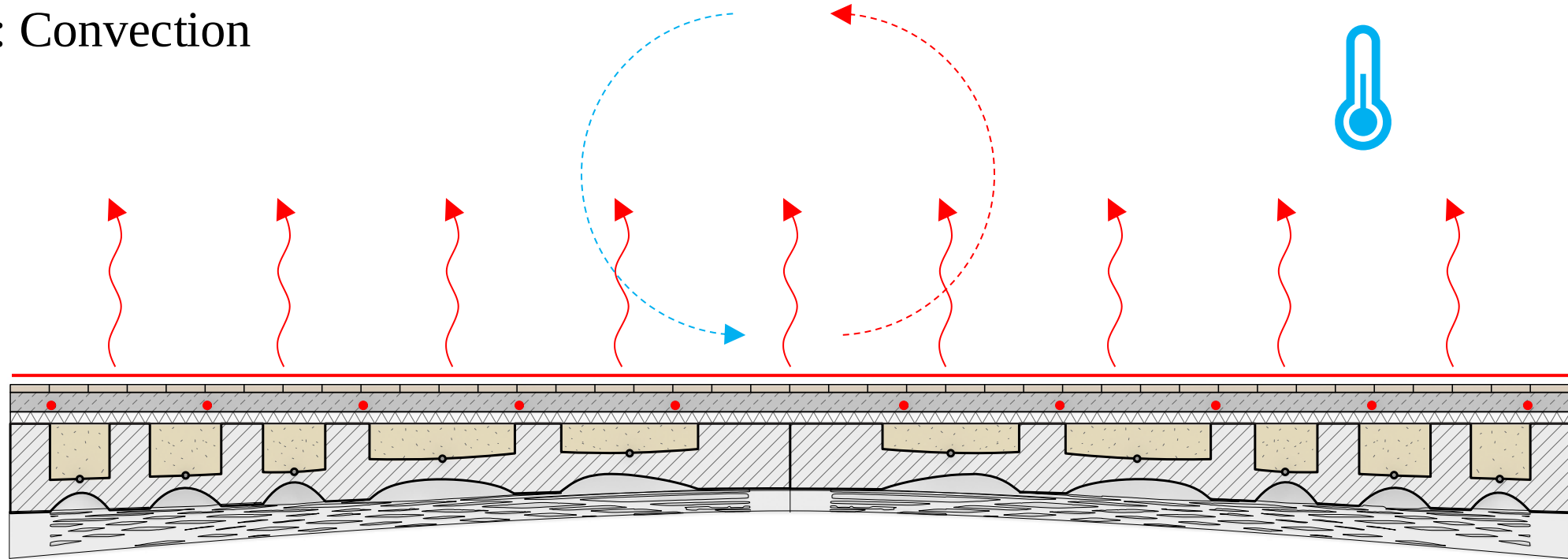
Heating: Floor Heating



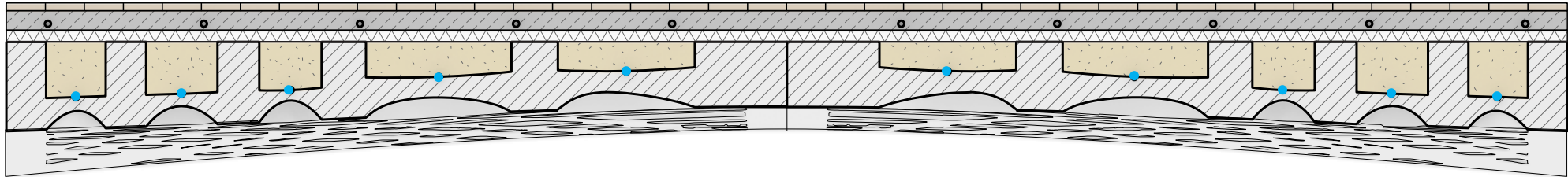
Heating: Radiation



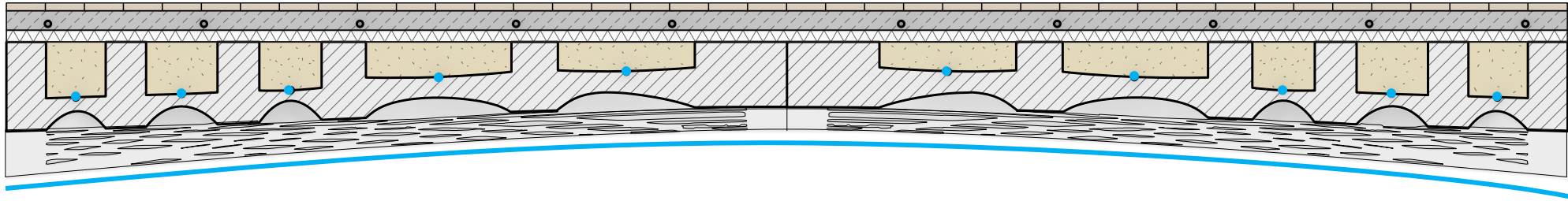
Heating: Convection



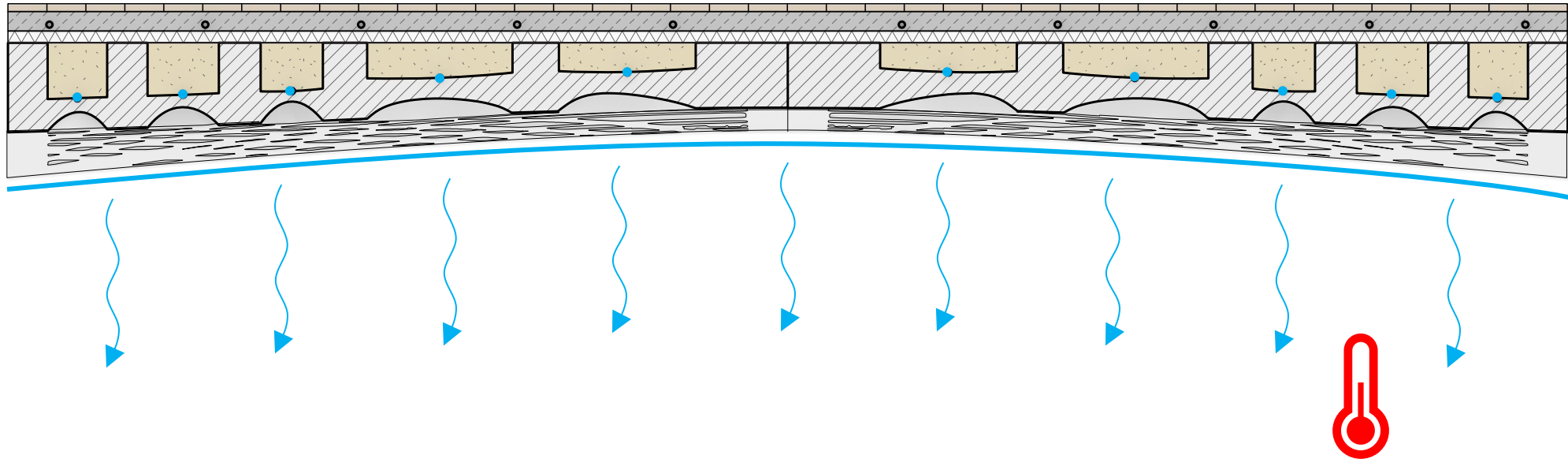
Cooling: Thermoactivated-Slab



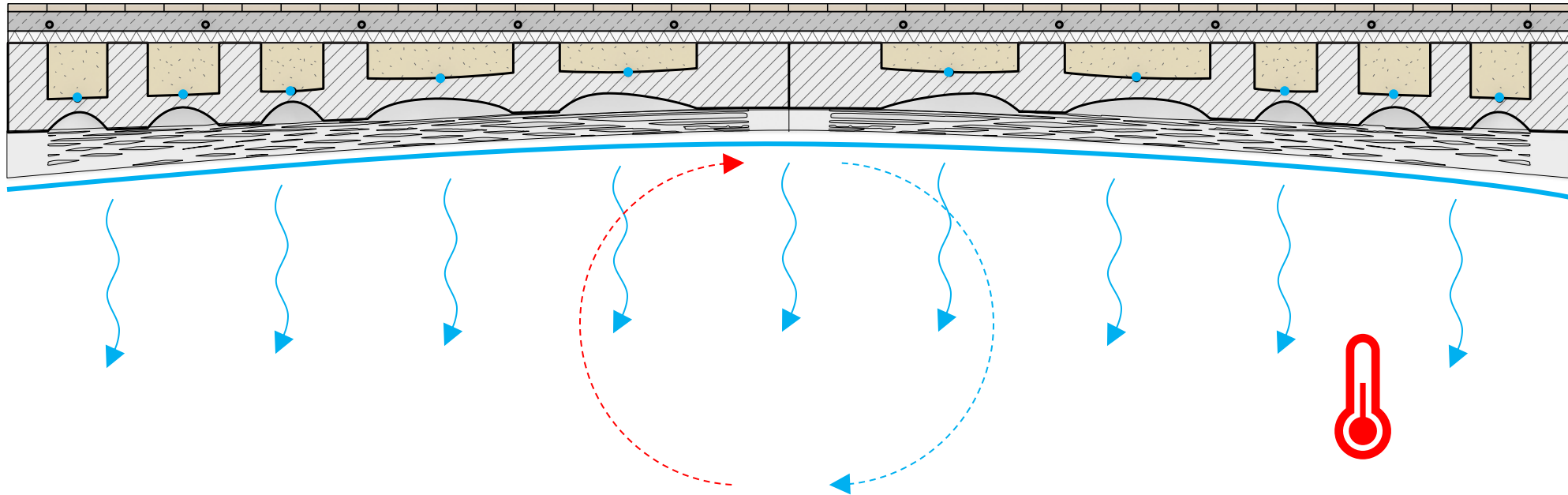
Cooling: Thermoactivated-Slab



Cooling: Radiation

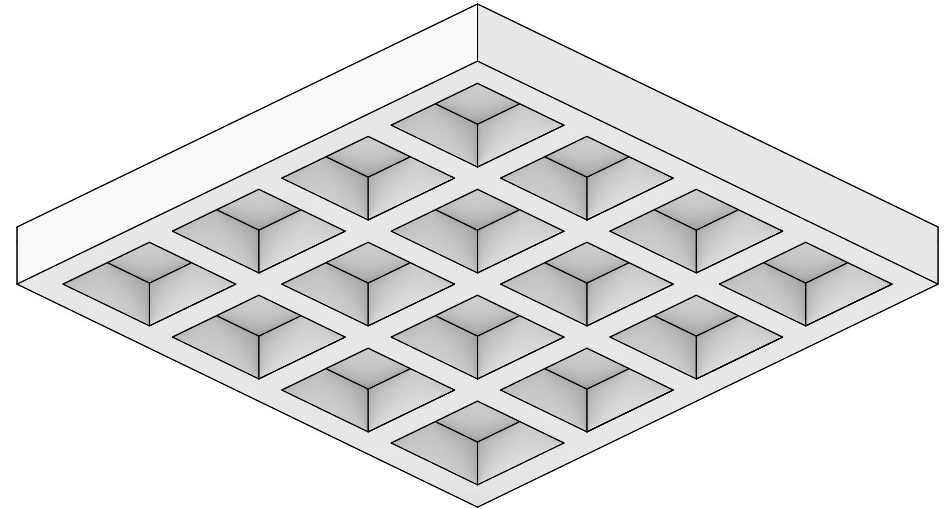
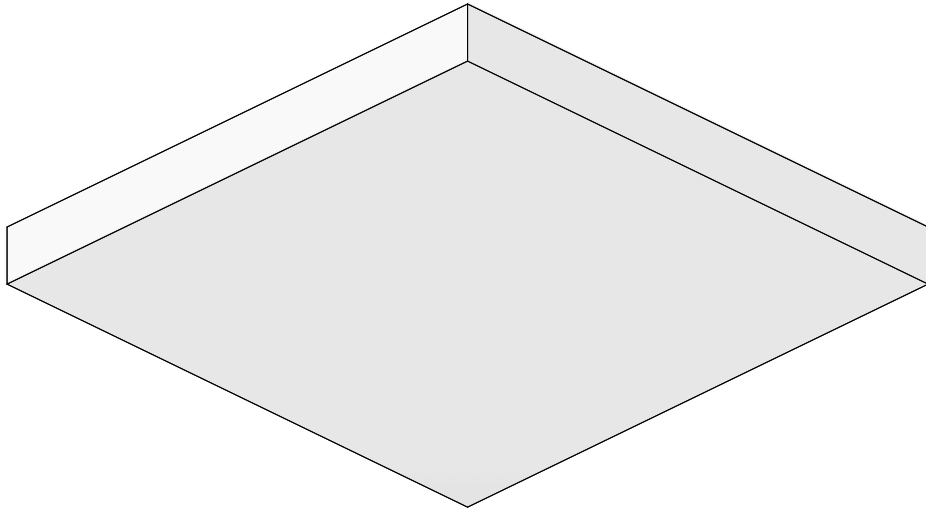


Cooling: Convection



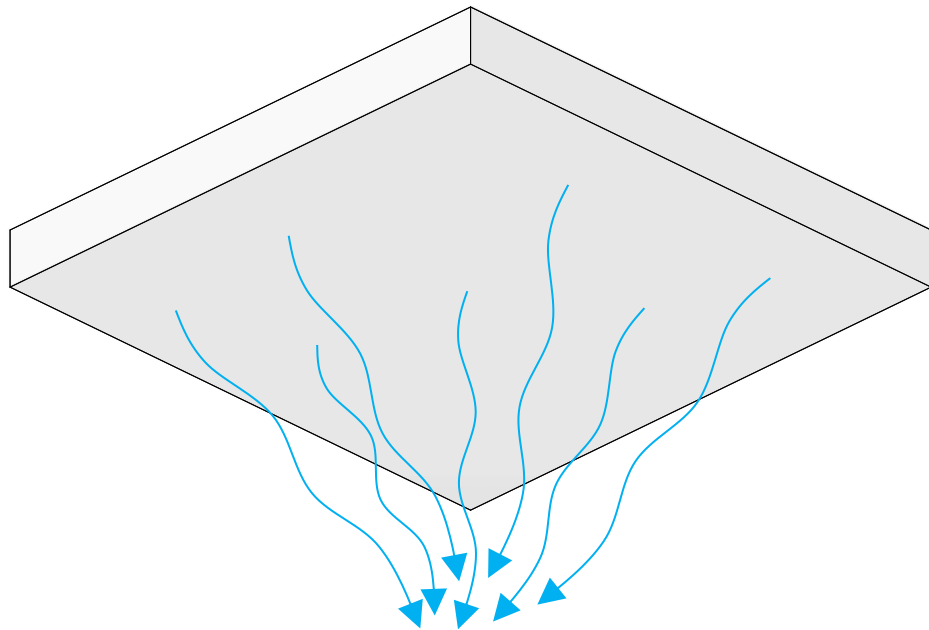
Slab Comparison

Flat Slab vs. Waffle Slab

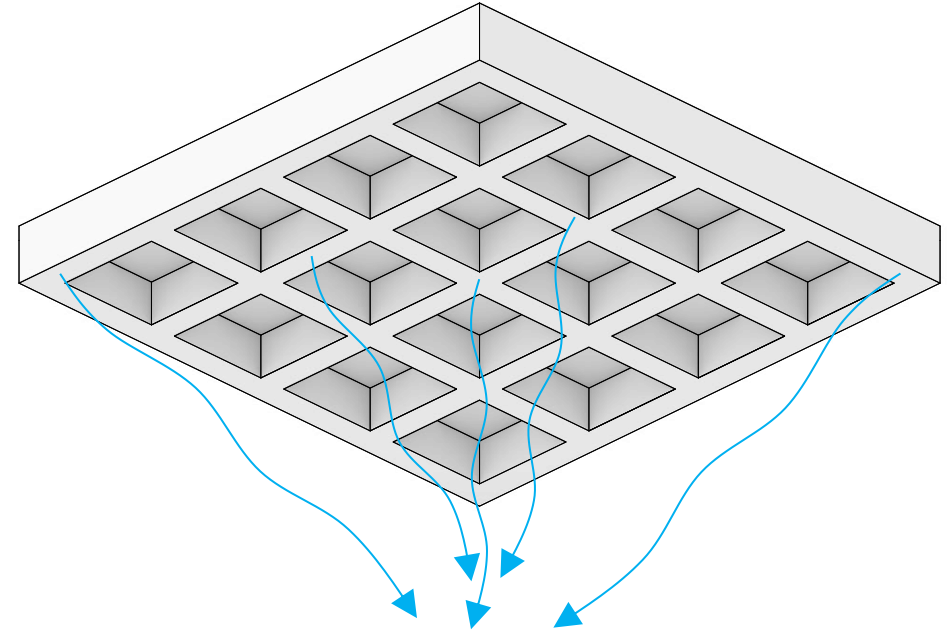


Cooling: Radiation Comparison

Flat Slab vs. Waffle Slab

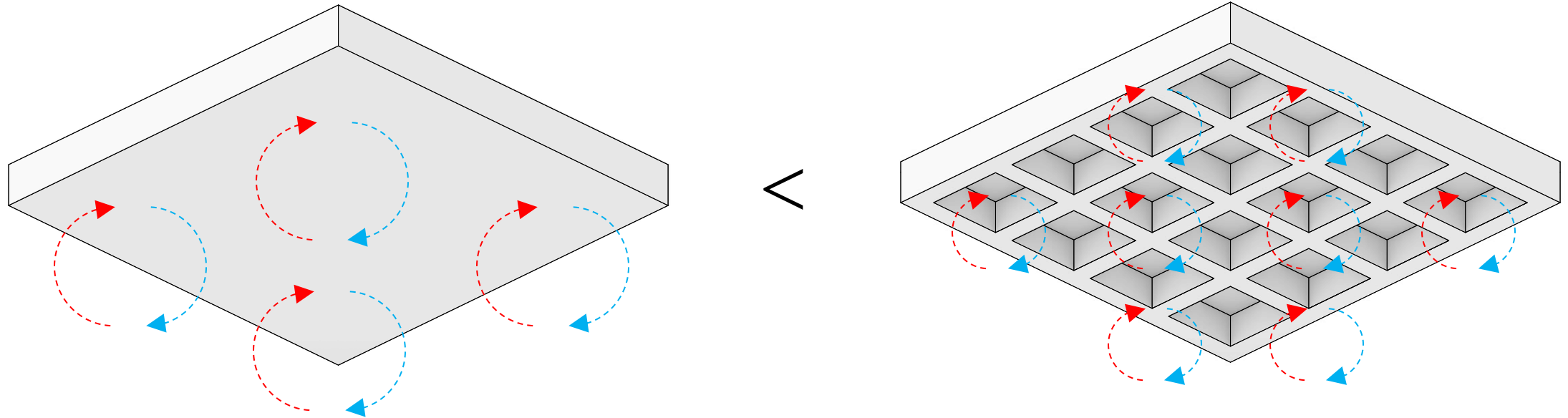


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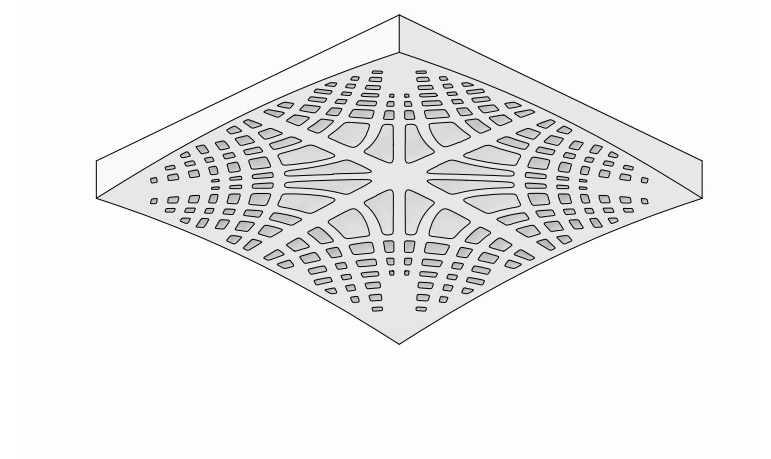
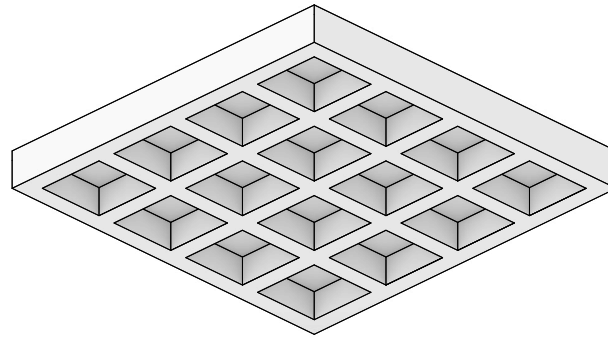
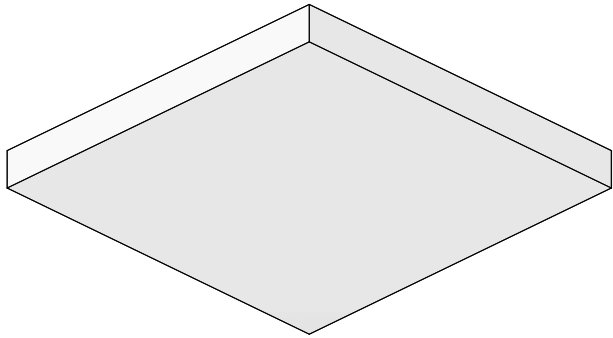
Cooling: Convection Comparison

Flat Slab vs. Waffle Slab



Slab Comparison

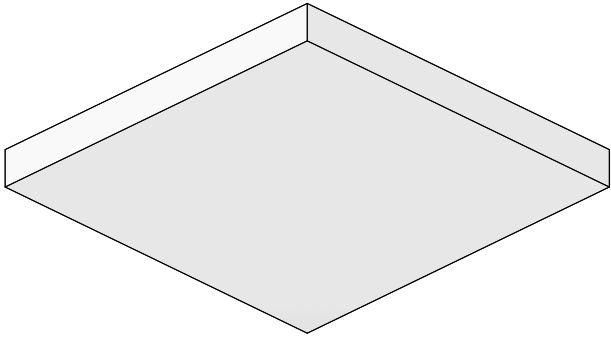
Flat Slab vs. Waffle Slab vs. Vault Slab



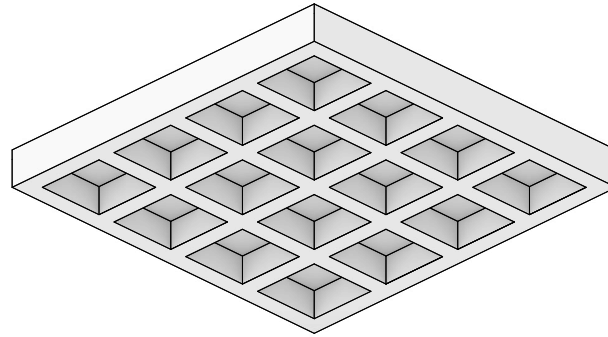
Surface Area Comparison

Flat Slab vs. Waffle Slab vs. Vault Slab

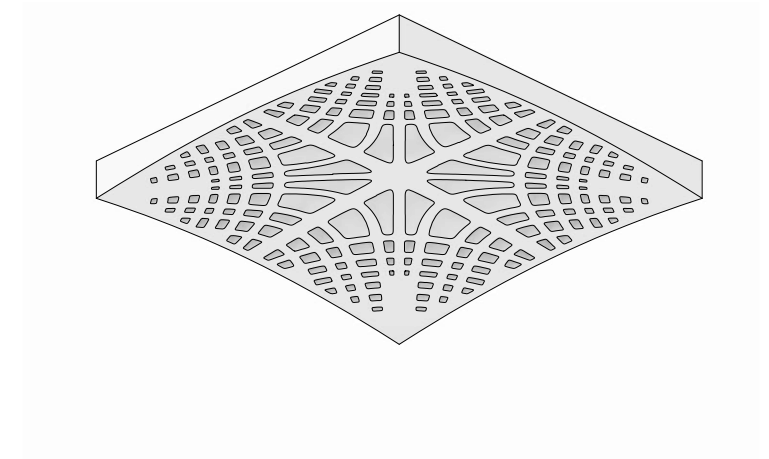
Surface Area: 16 m²



Surface Area: 30 m²



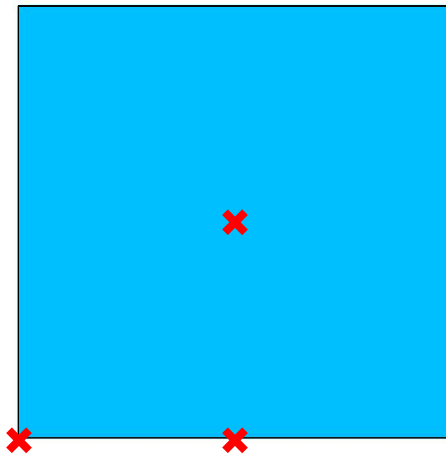
Surface Area: 18 m²



Visibility Percentage Comparison

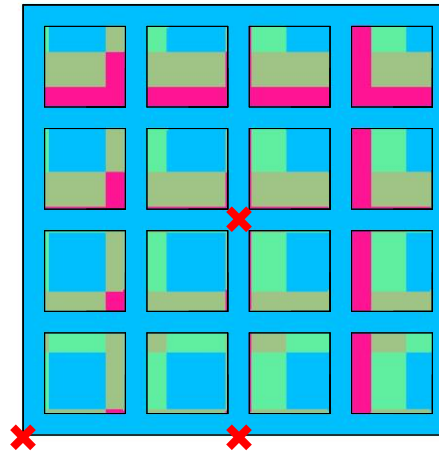
Flat Slab vs. Waffle Slab vs. Vault Slab

Surface Area: 16 m²



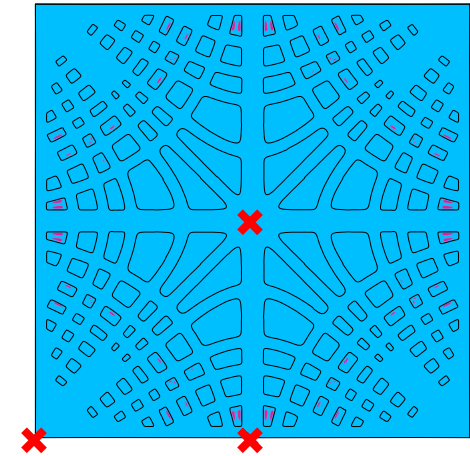
Visible Surface Area: 16 m²

Surface Area: 30 m²



Visible Surface Area: 8 – 17 m²

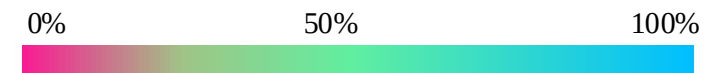
Surface Area: 18 m²



Visible Surface Area: 17 – 18 m²

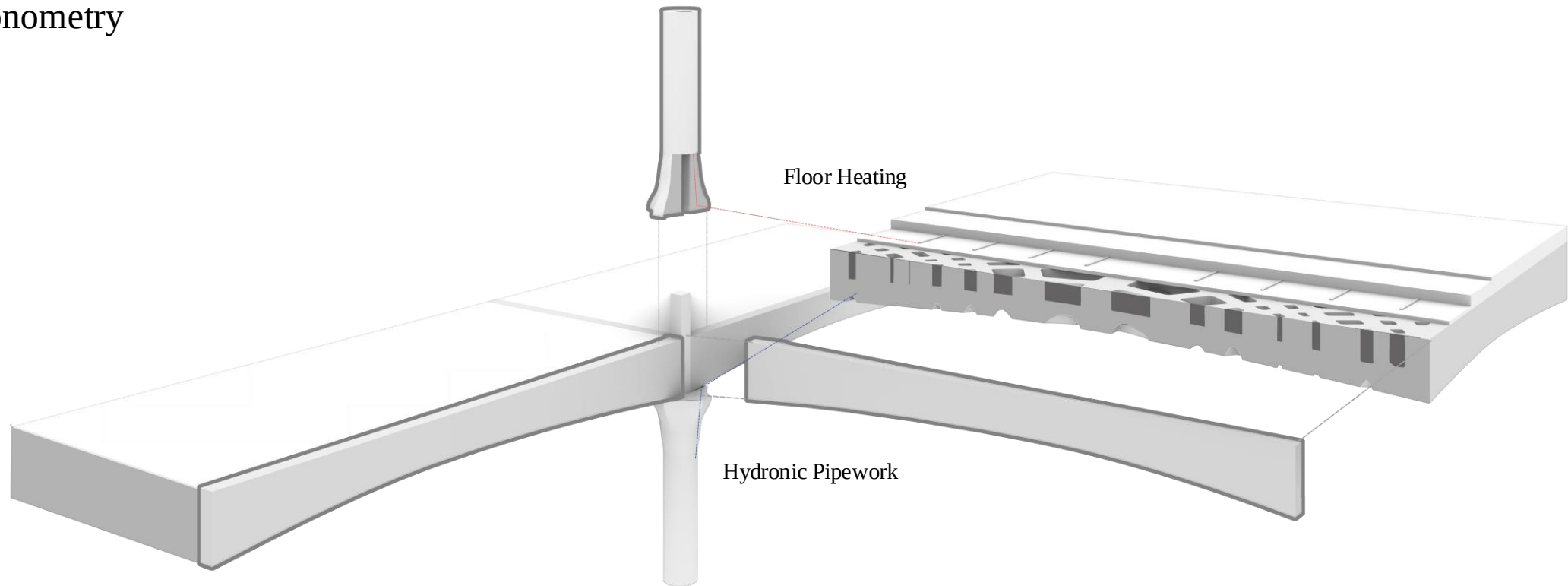
Measurement positions 
1,5 m below the slab

Visibility Percentage
According to Ladybug Tools*



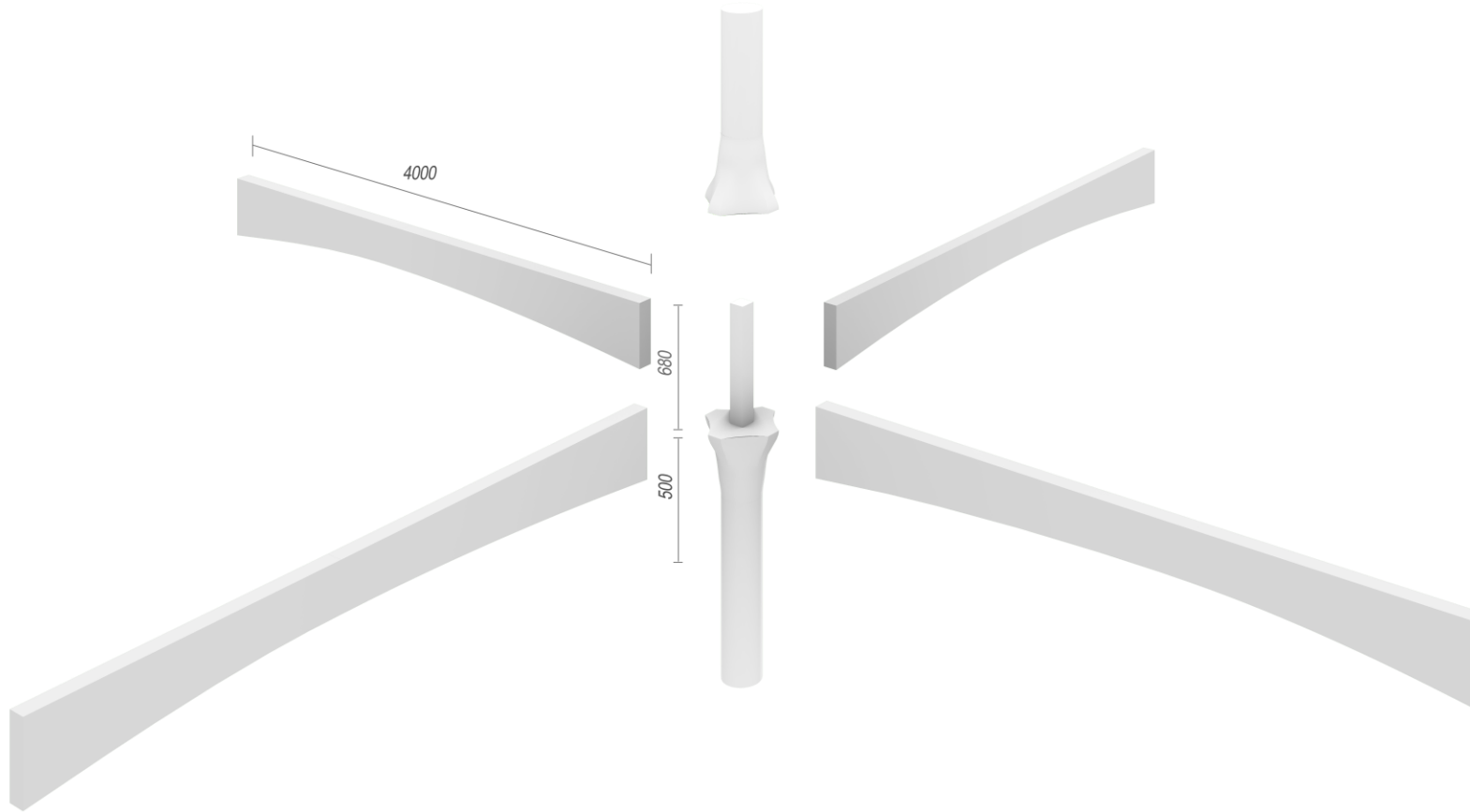
Introduction

Joint Isonometry



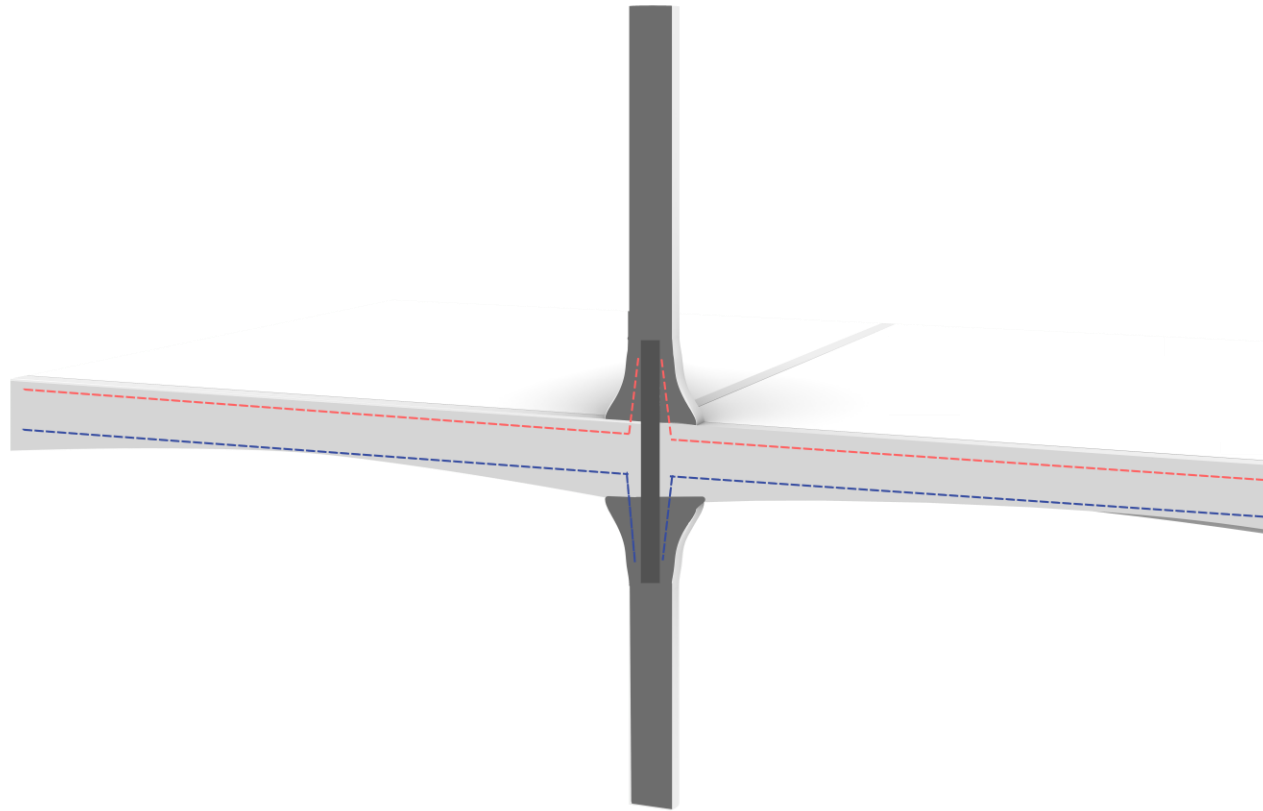
Introduction

Joint



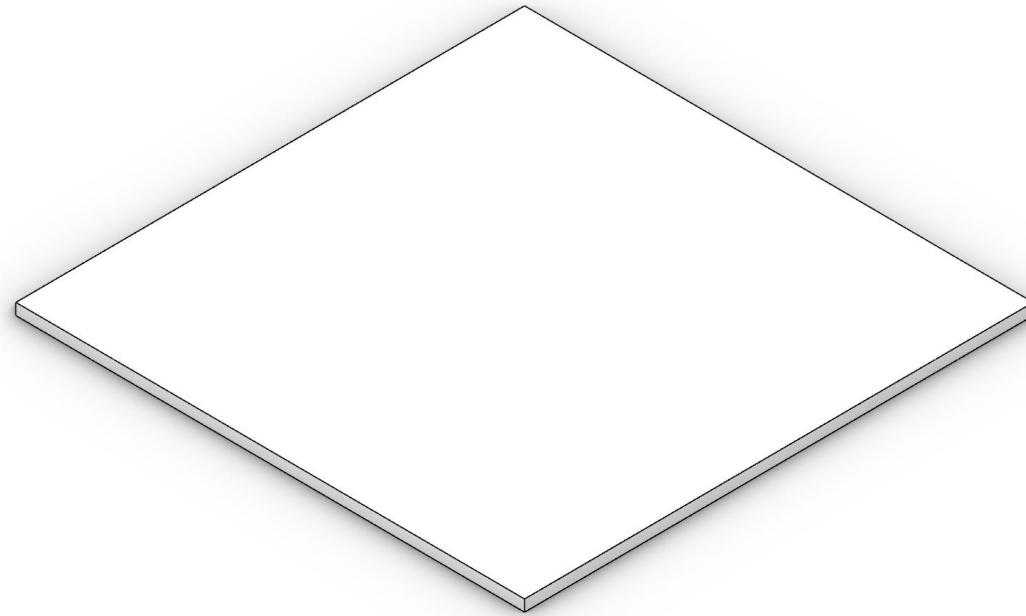
Introduction

Joint- column



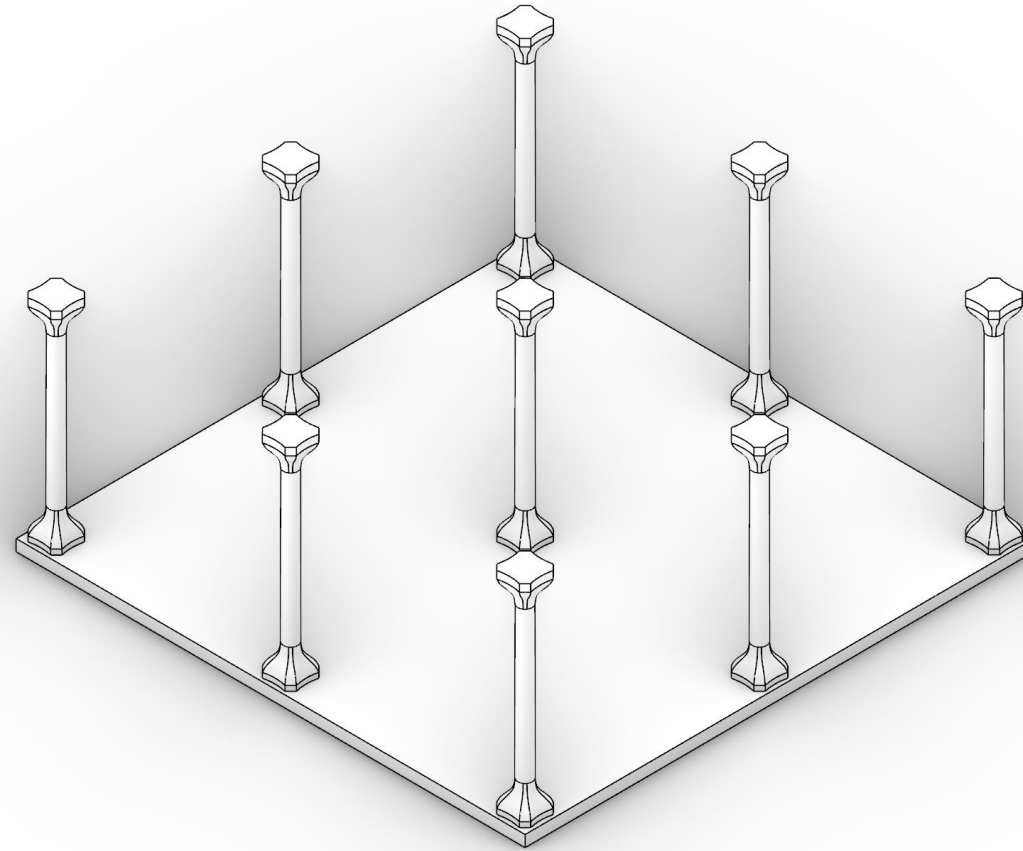
Construction Process

1. Base



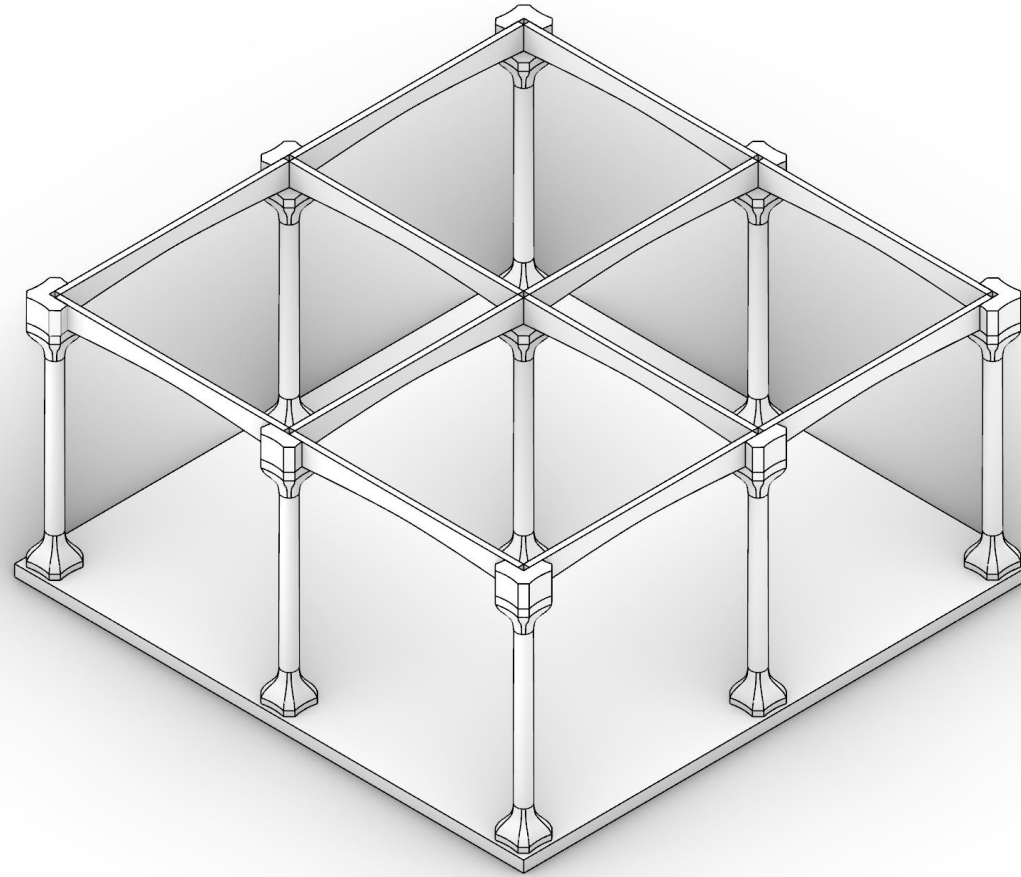
Construction Process

2. Columns



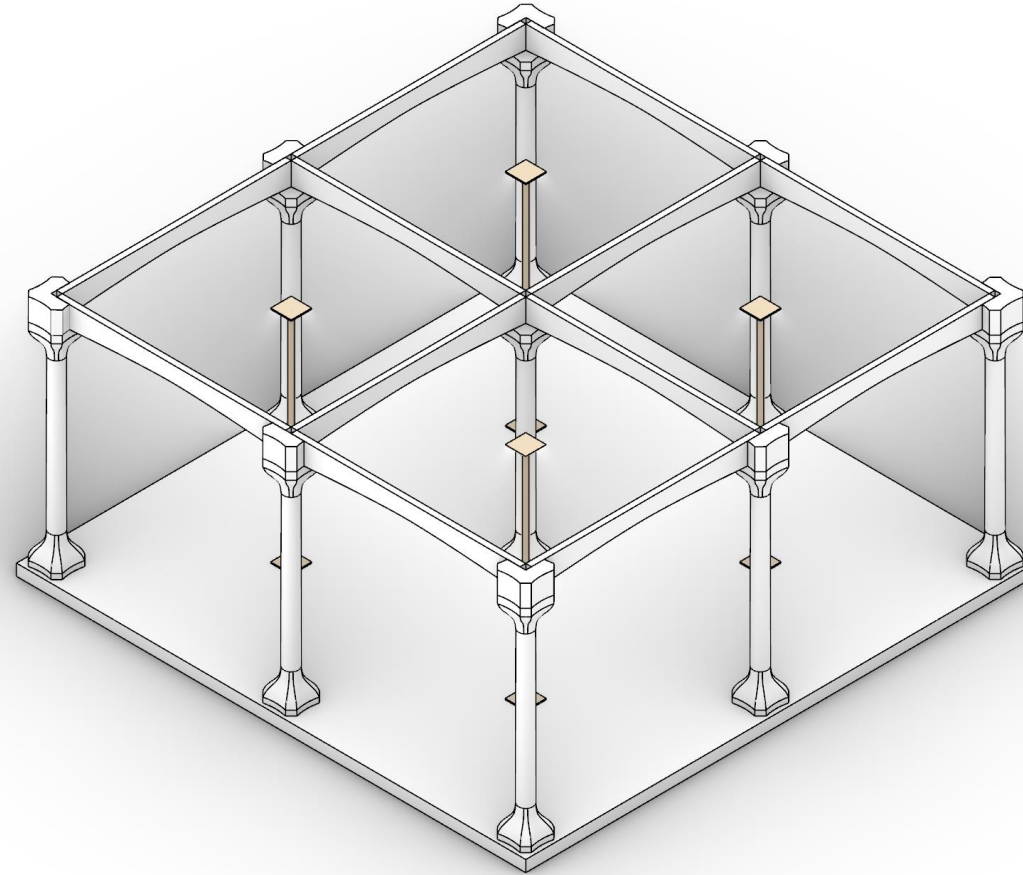
Construction Process

3. Arched Frames



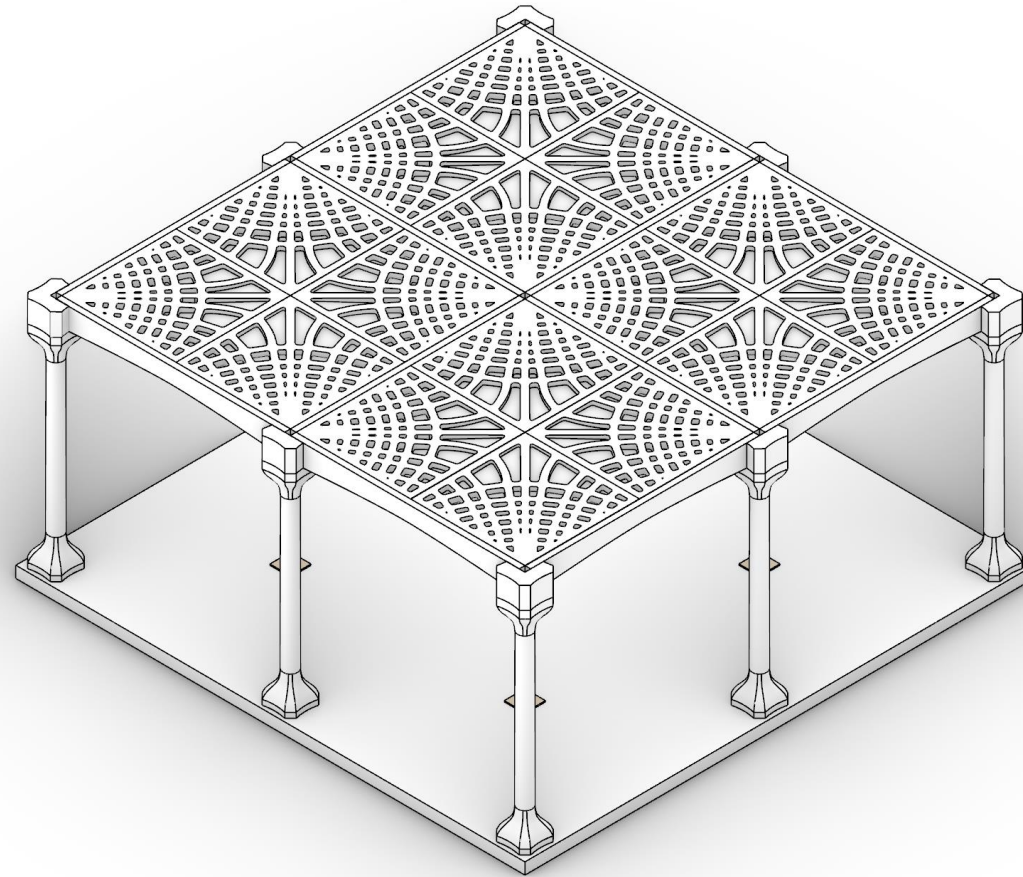
Construction Process

4. Central Support



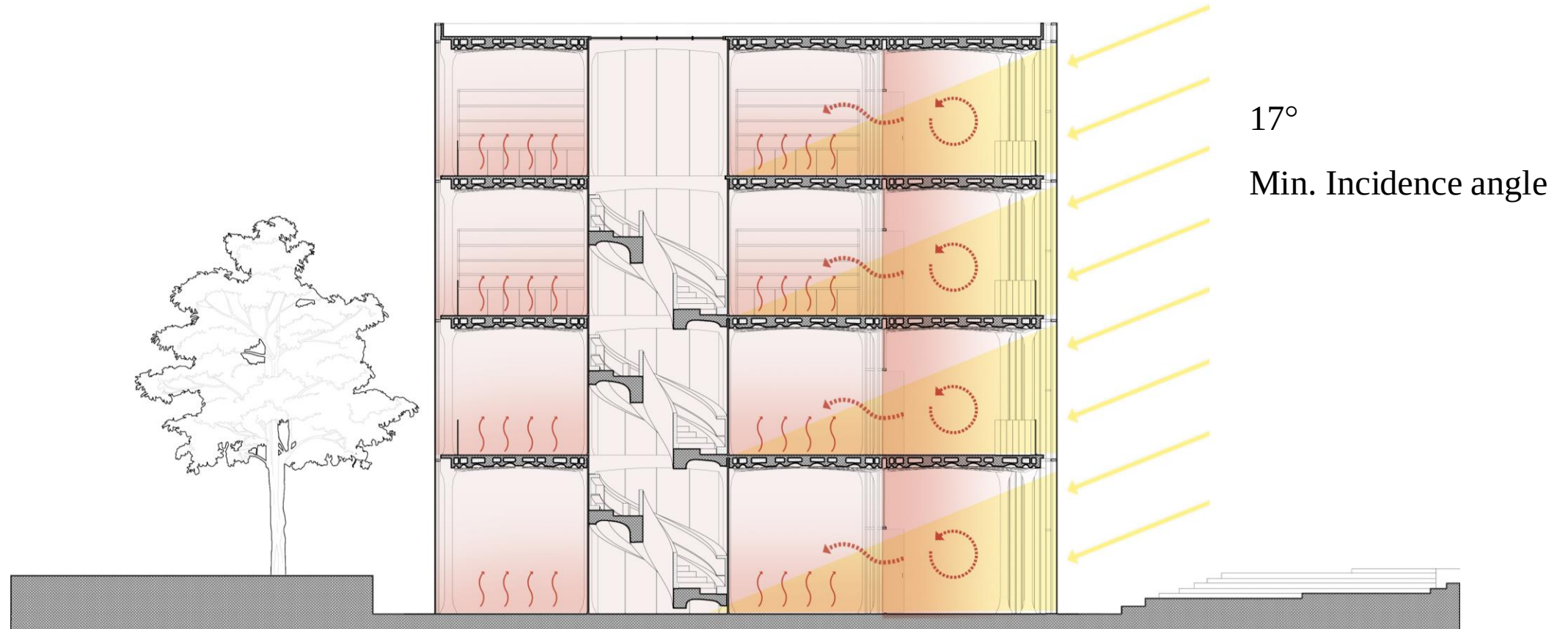
Construction Process

5. Add Vaults



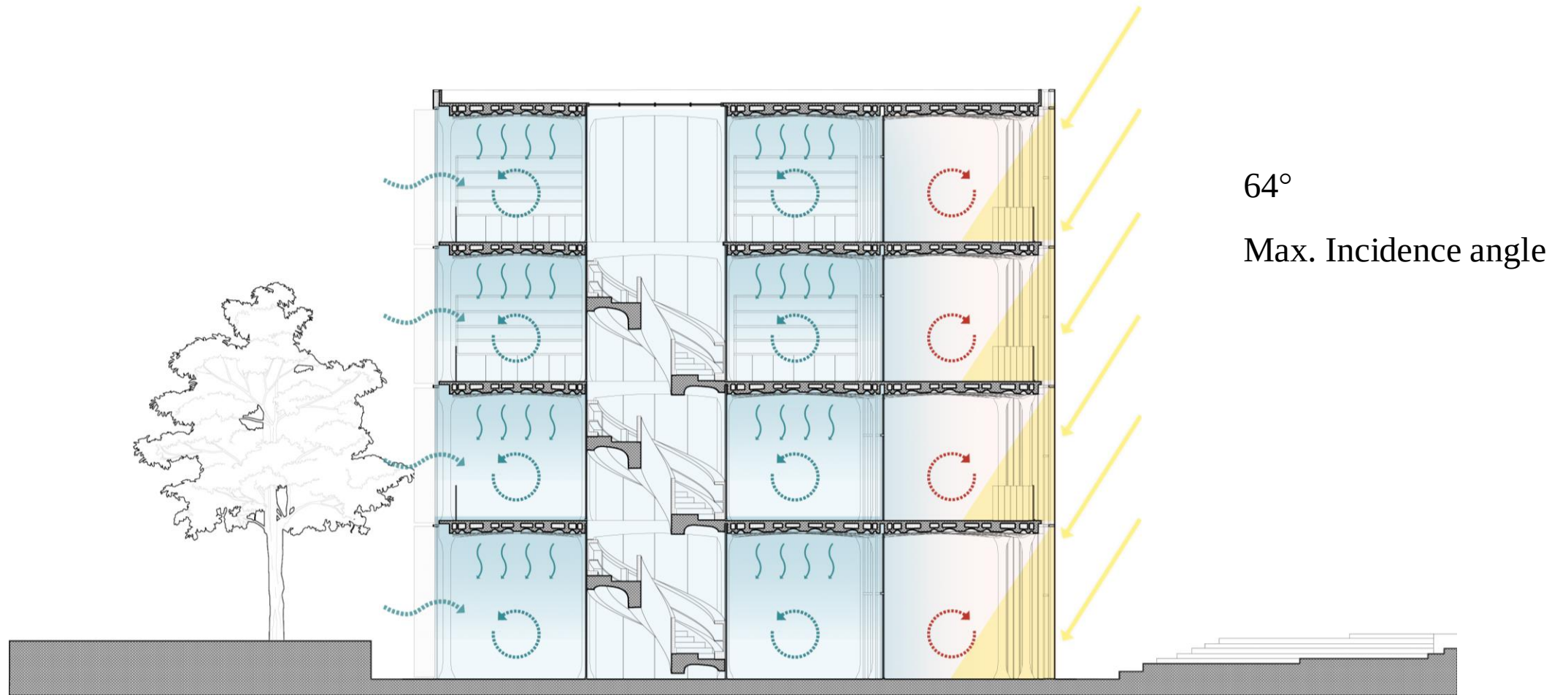
Climate concept

Winter

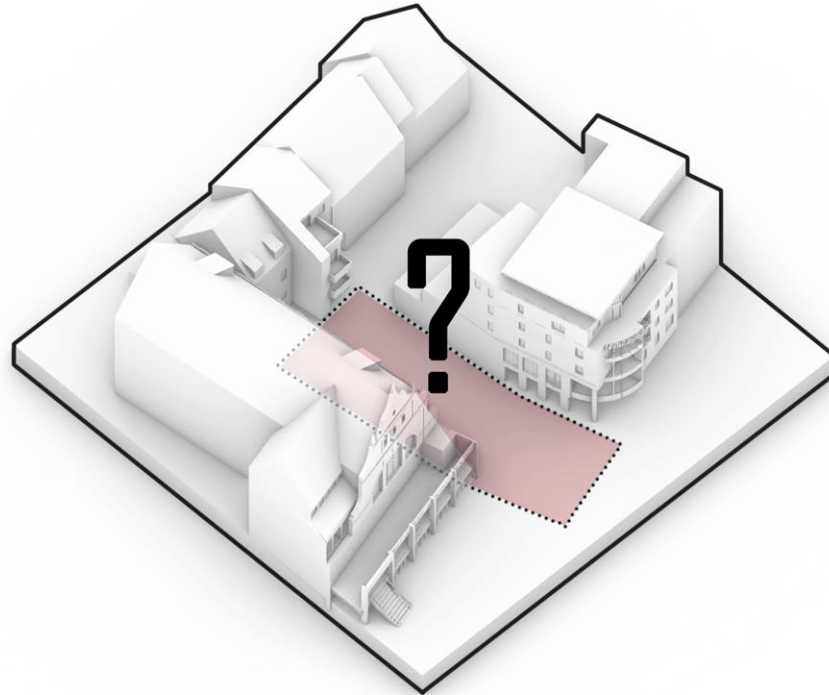


Climate concept

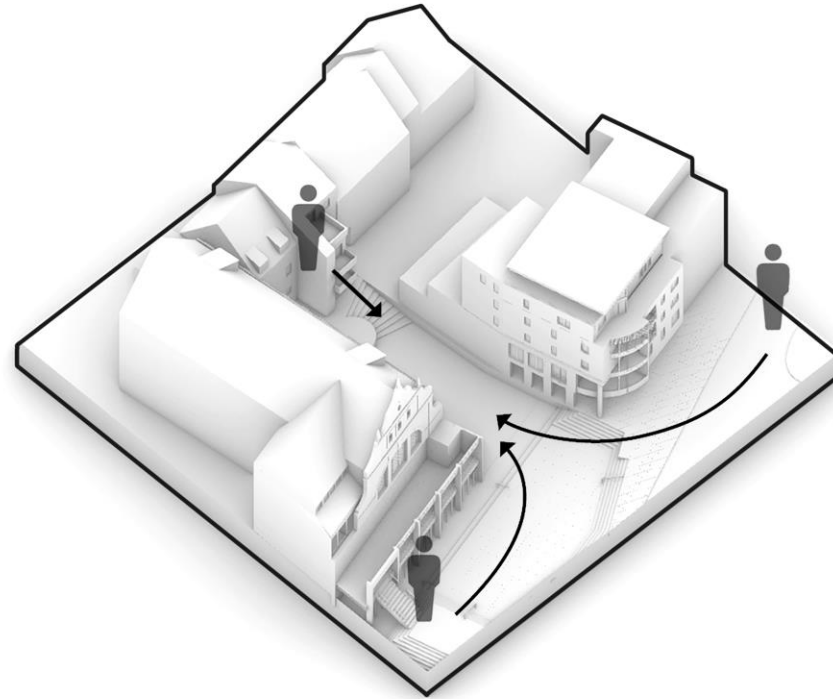
Summer



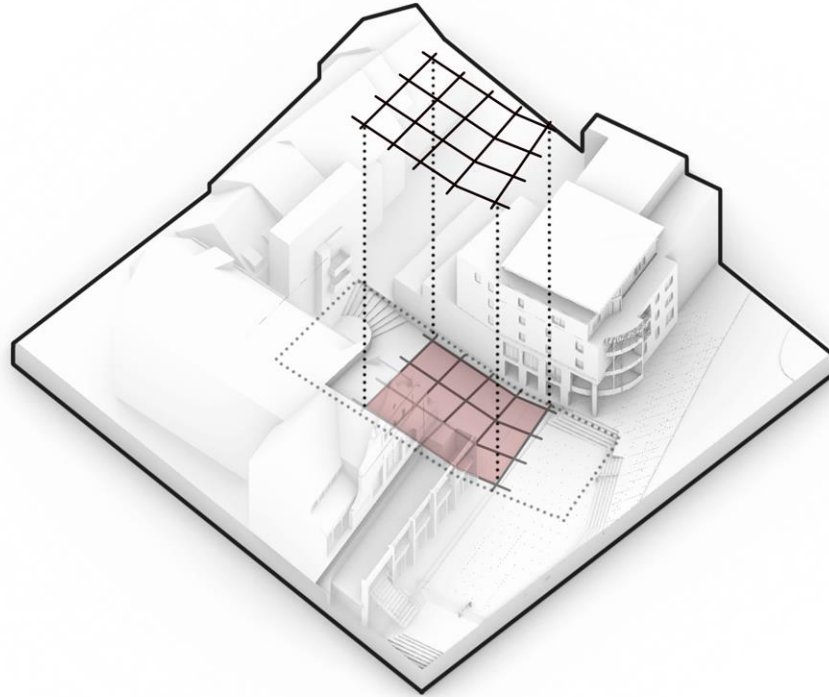
The Site

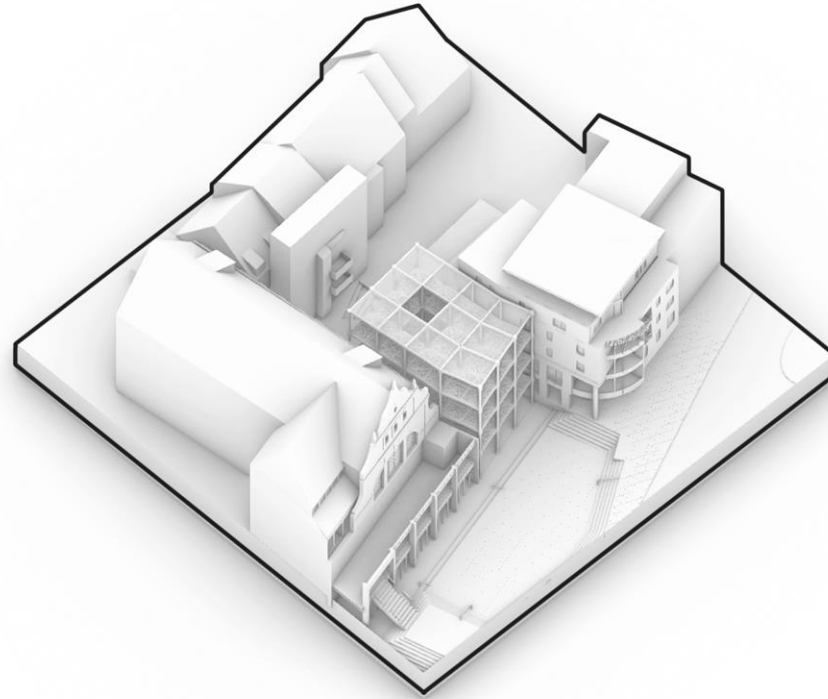


Access

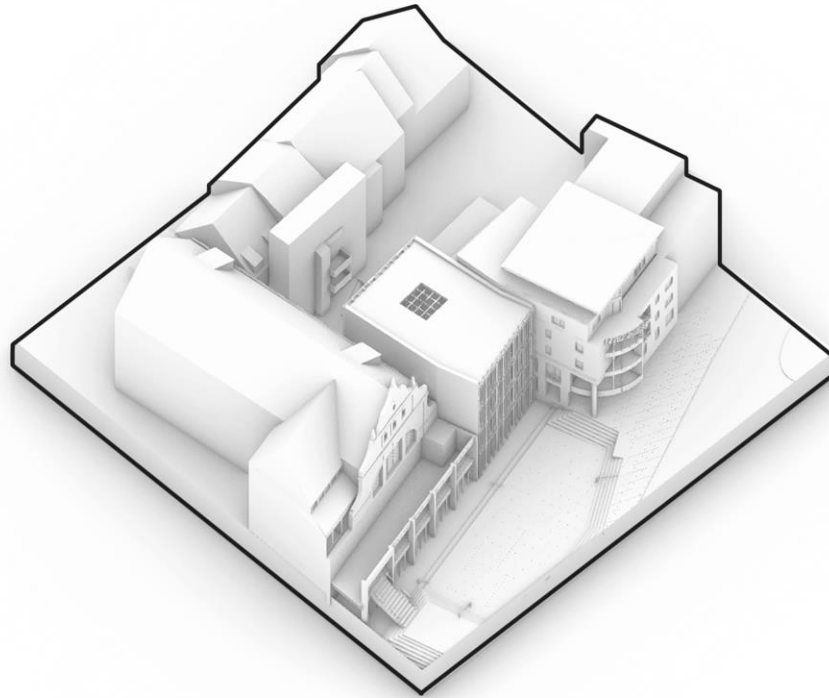


Grid





Envelope



Site Plan



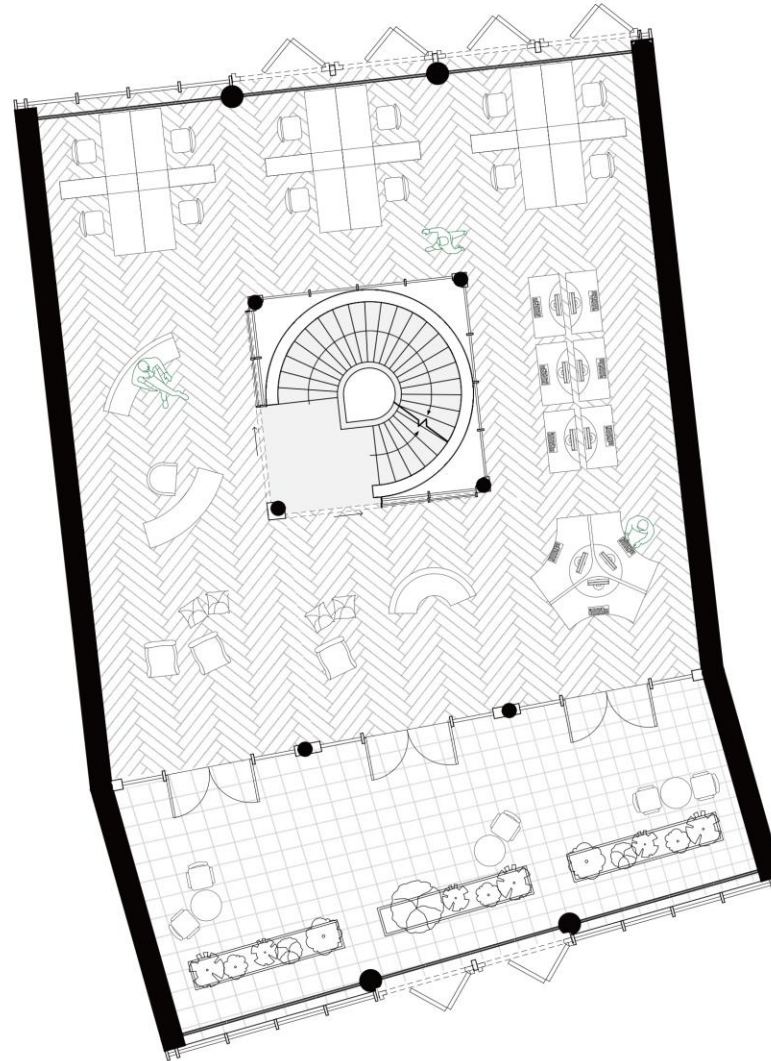
Floor plan

Ground floor



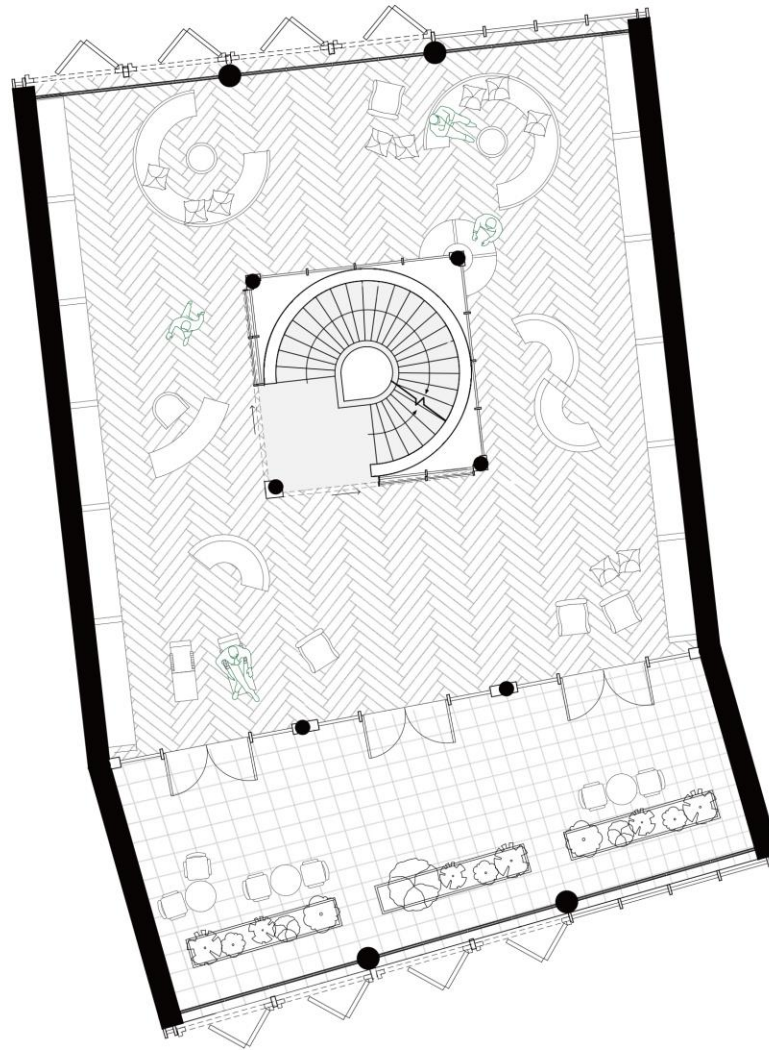
Floor plan

First floor



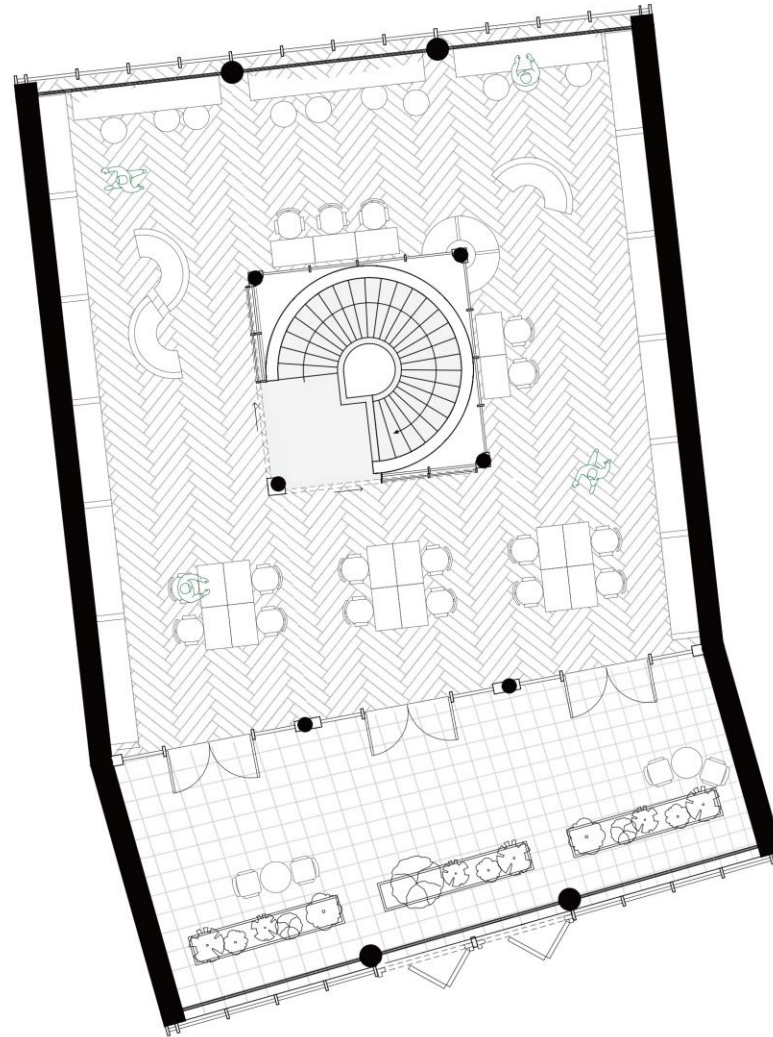
Floor plan

Second floor



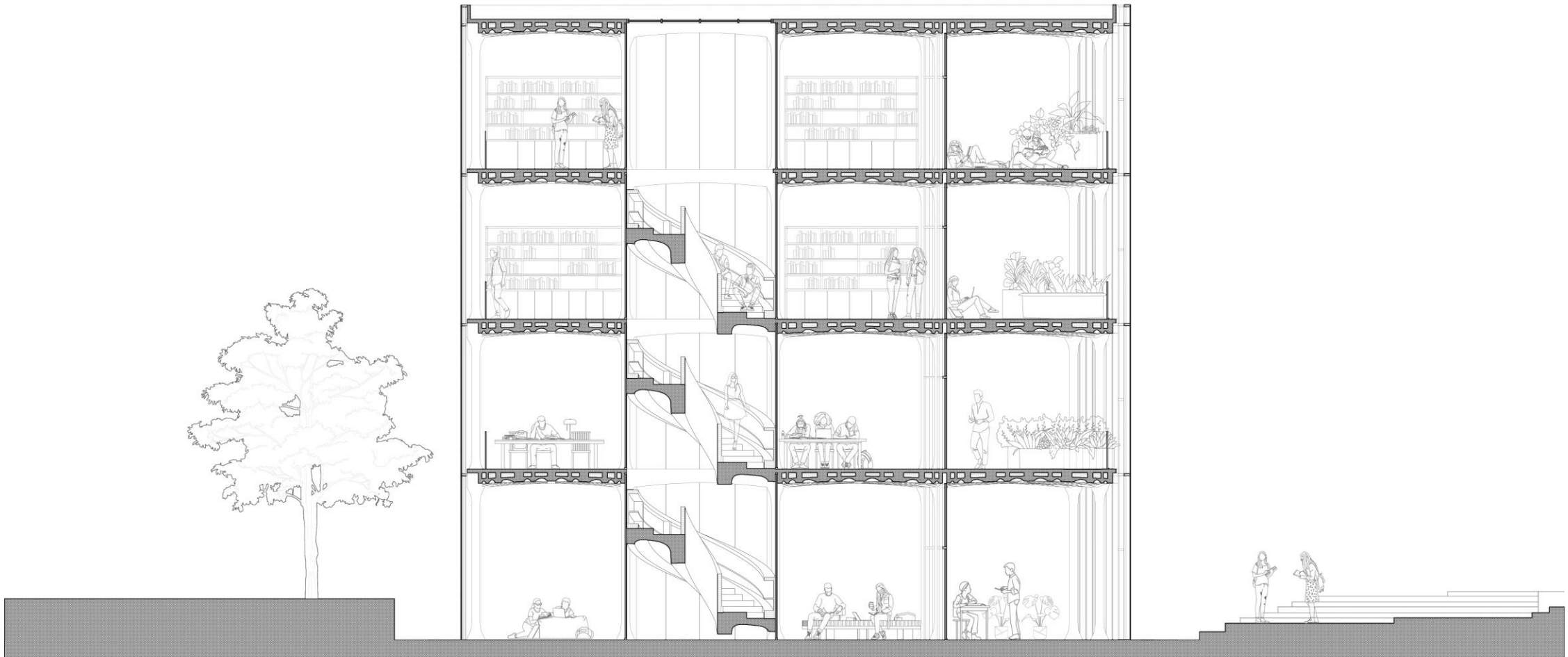
Floor plan

Third floor



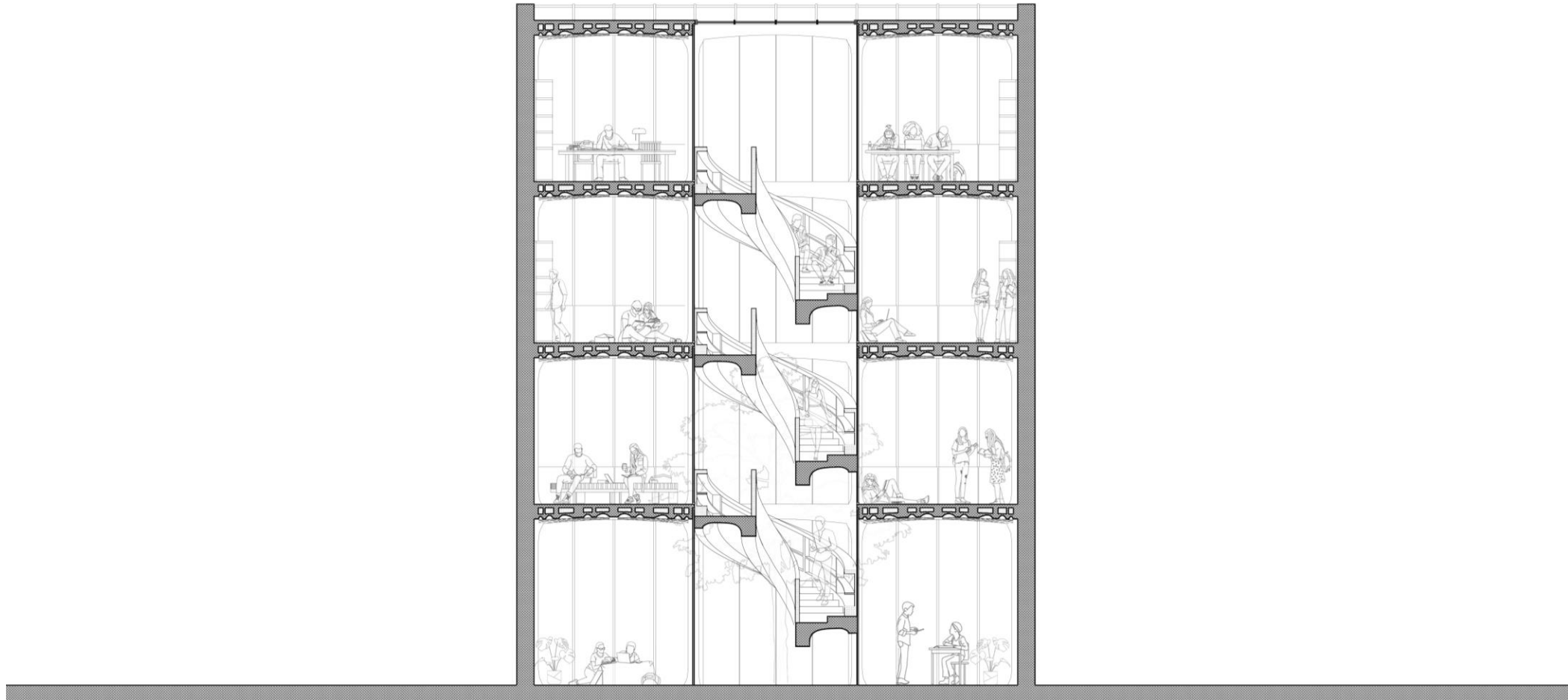
Section

Cross Section



Section

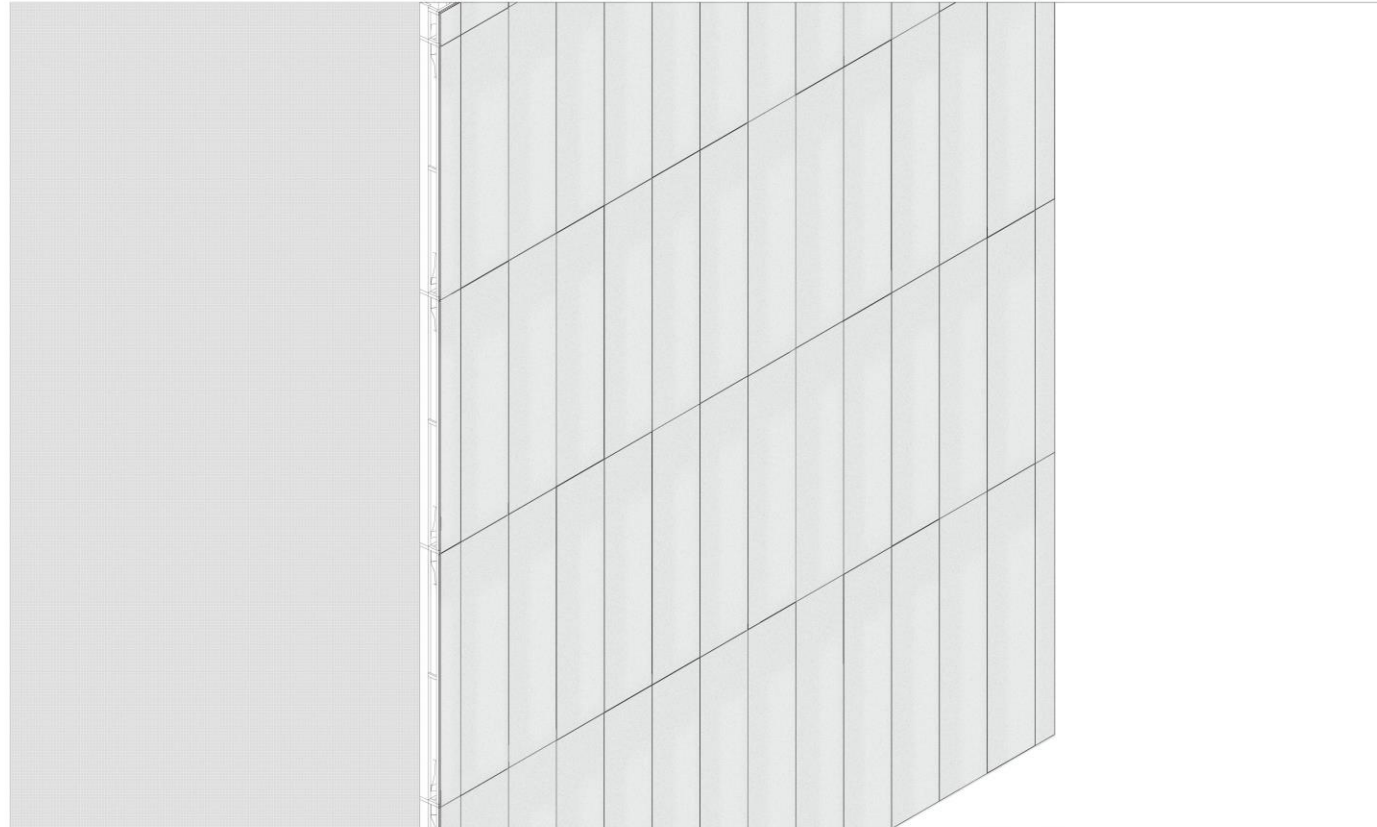
Cross Section



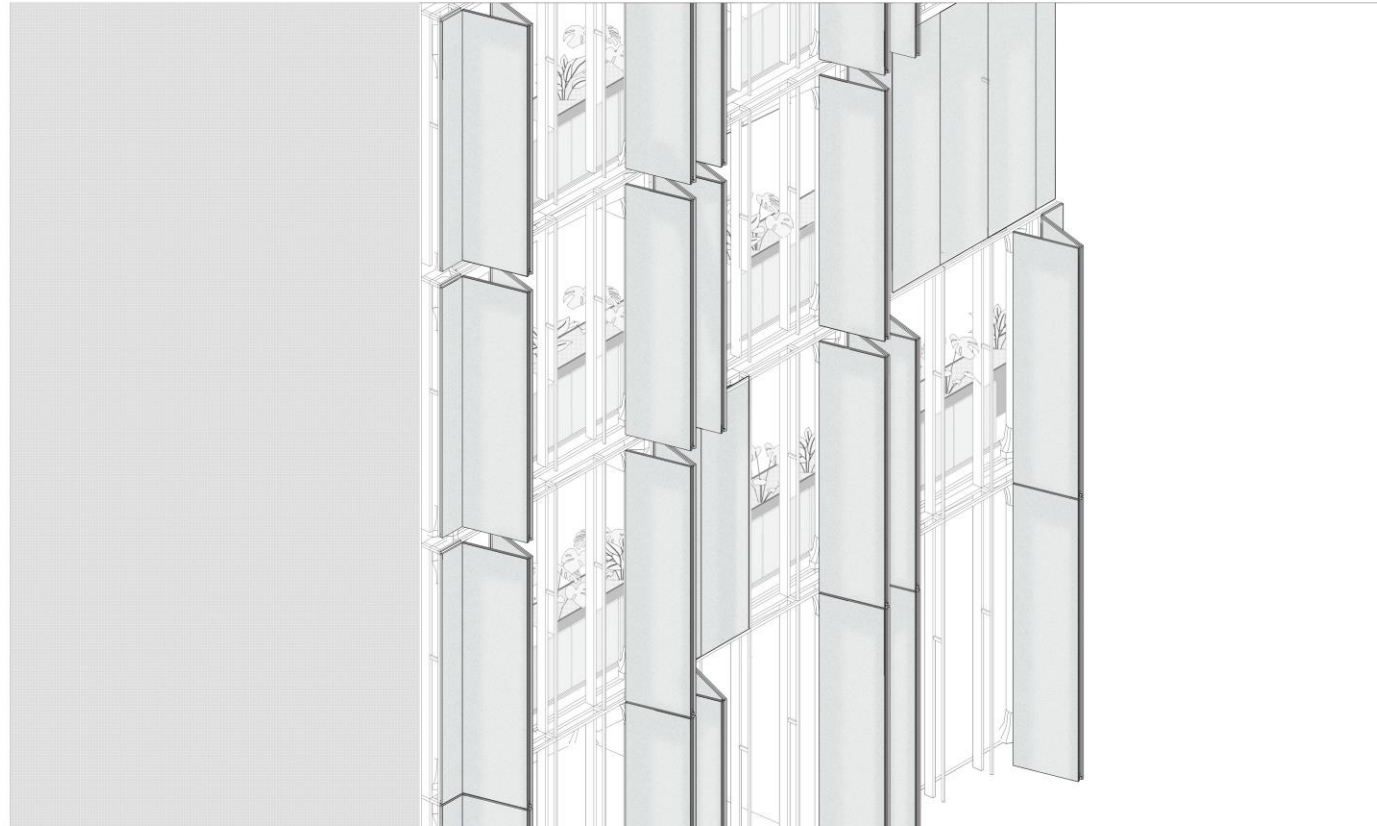
Facade



Façade System



Façade System



Exterior Visualization



Interior Visualization: Winter Garden



Interior Visualization



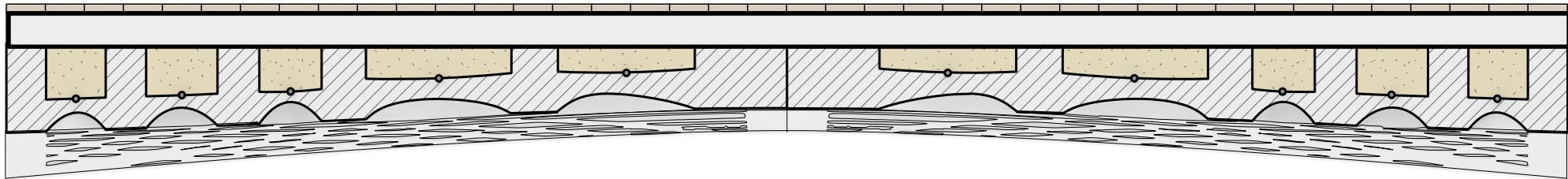
Exterior Visualization



Thank You

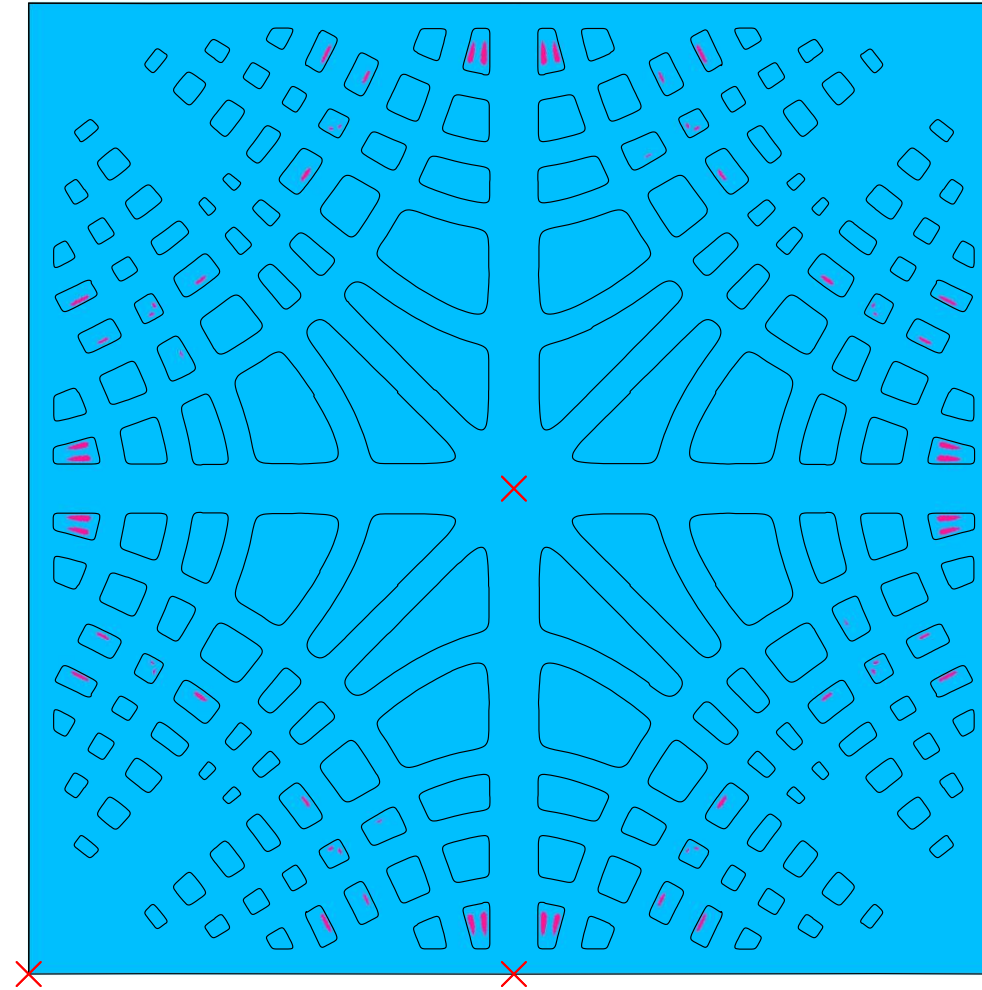
Bonus: Critical Reflection

Why add screed and insulation?

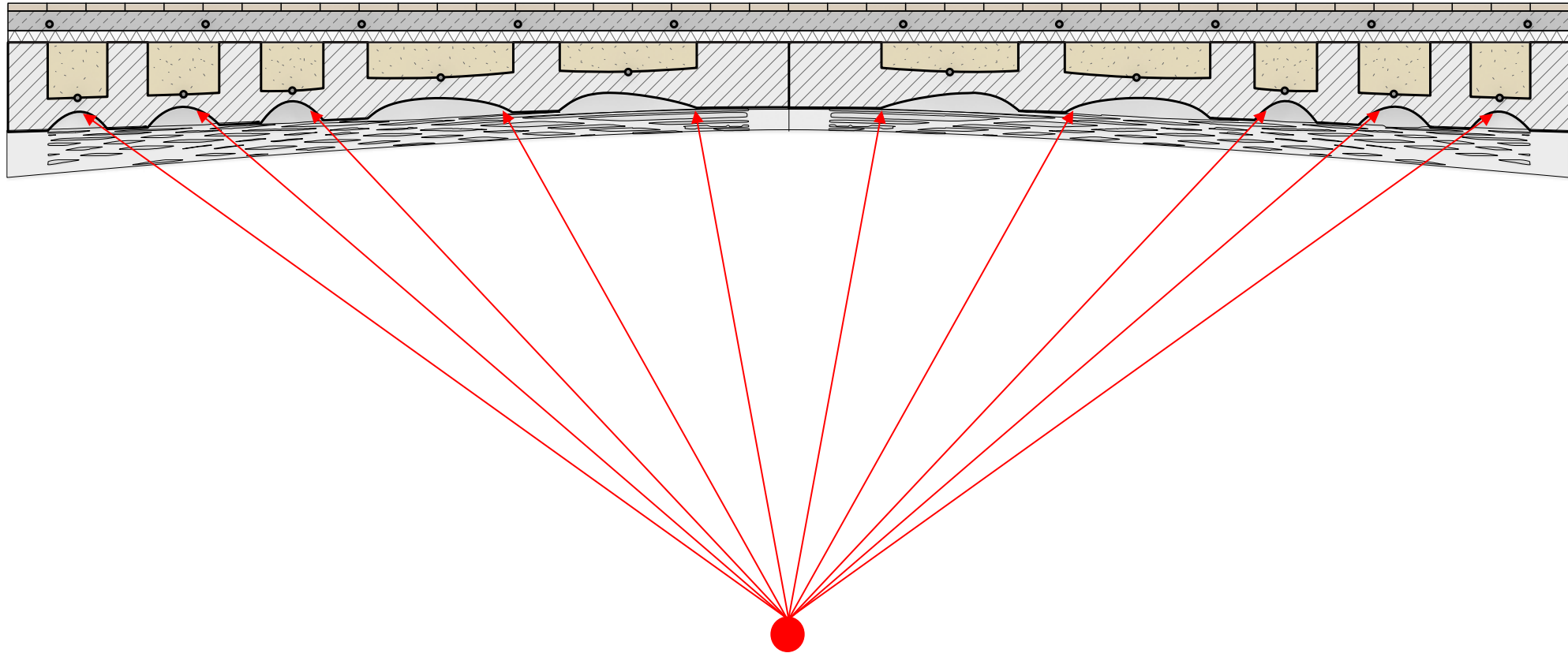


- ✗ Unpractical for flooring
- ✗ No separate systems for heating and cooling
- ✗ Structure Borne Noise Transmission
- ✗ Installation of the Hydronic System

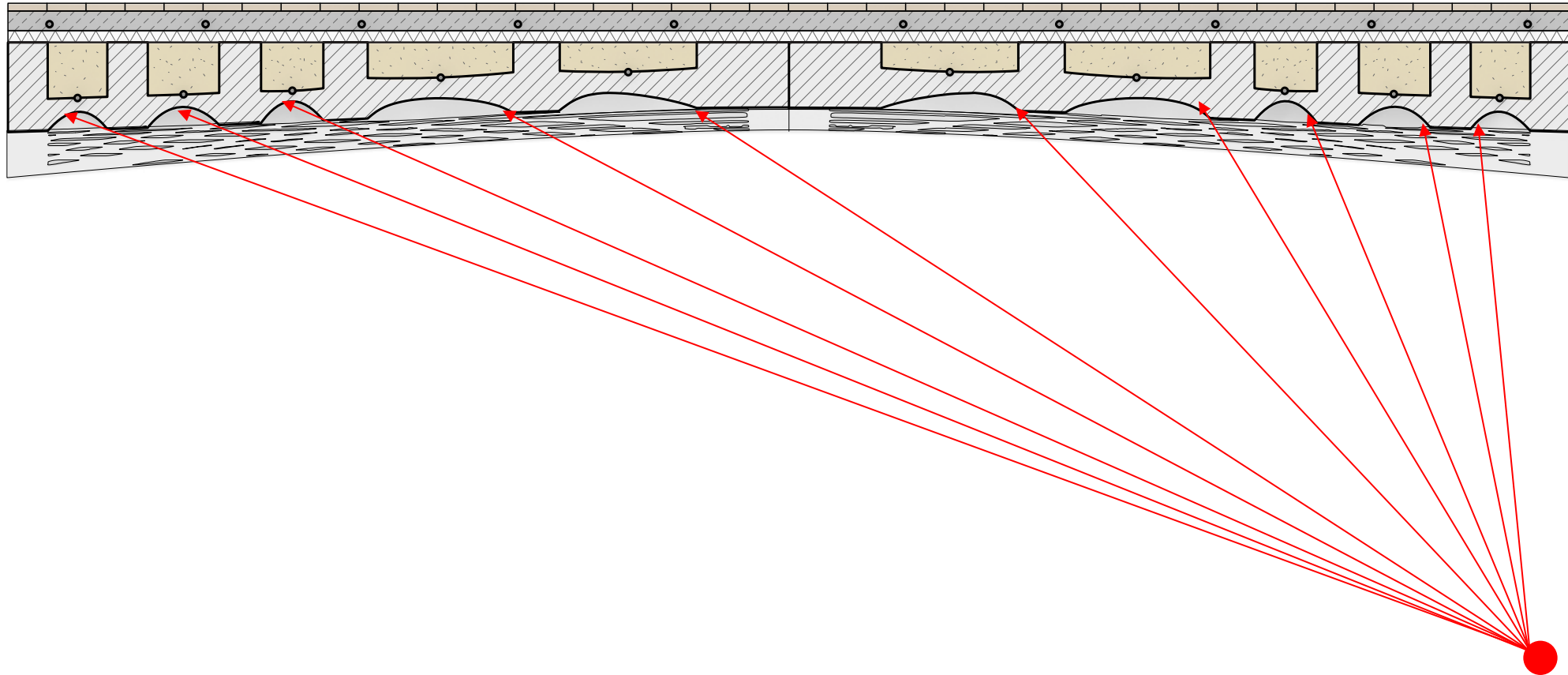
Visibility Factor



Visibility Factor



Visibility Factor



Facade



Bracing

